

BFS Traversal

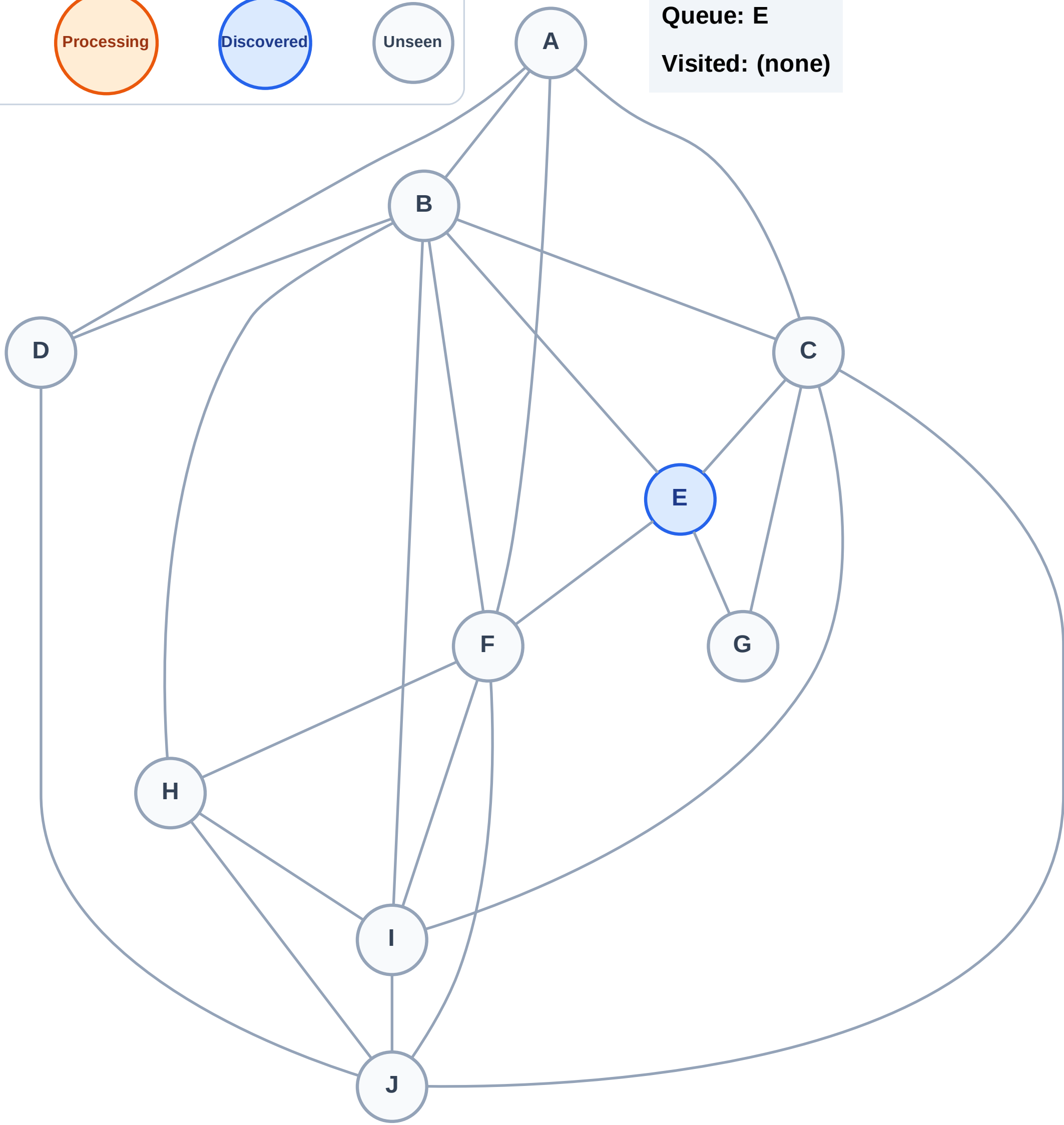
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

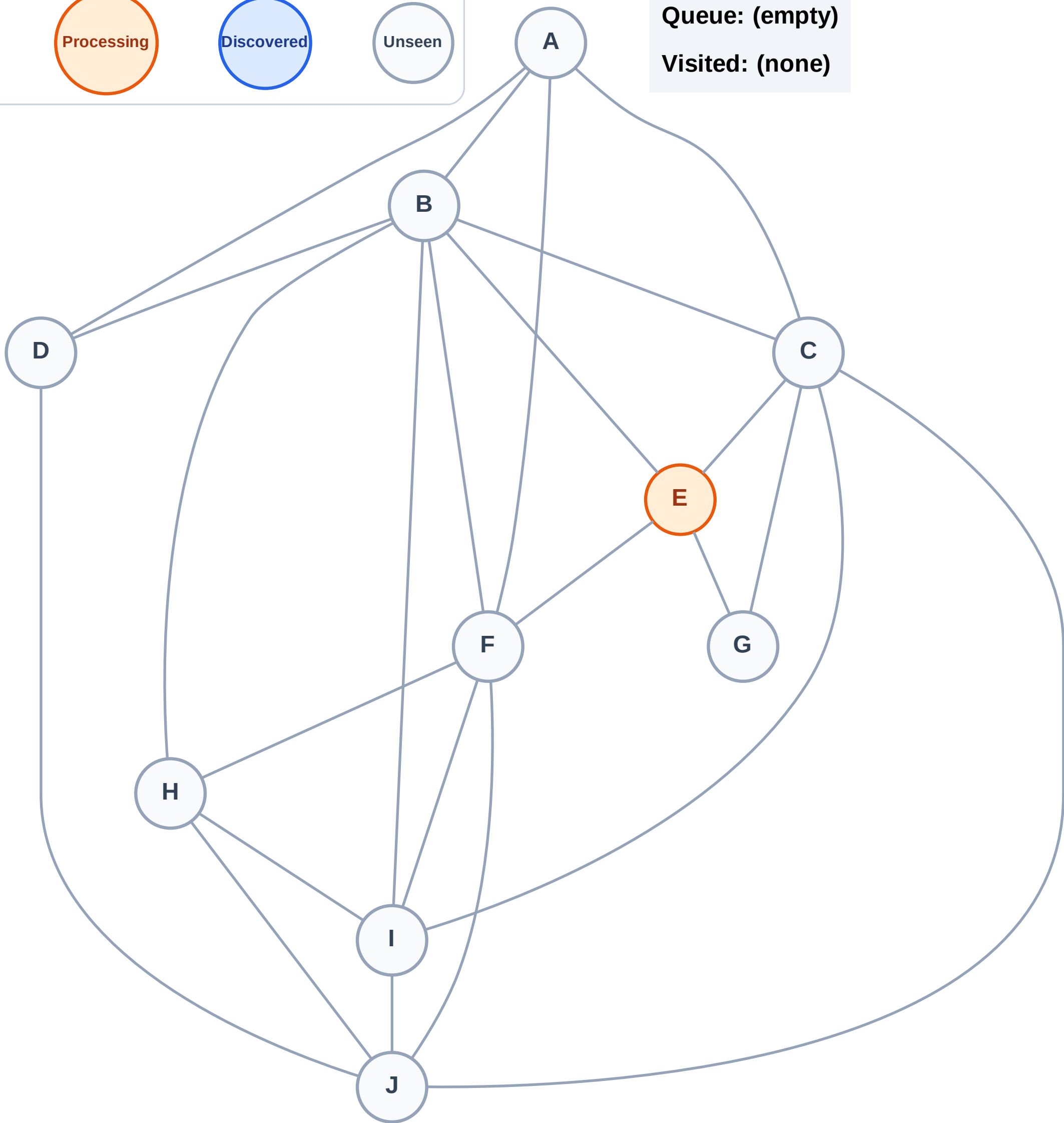
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

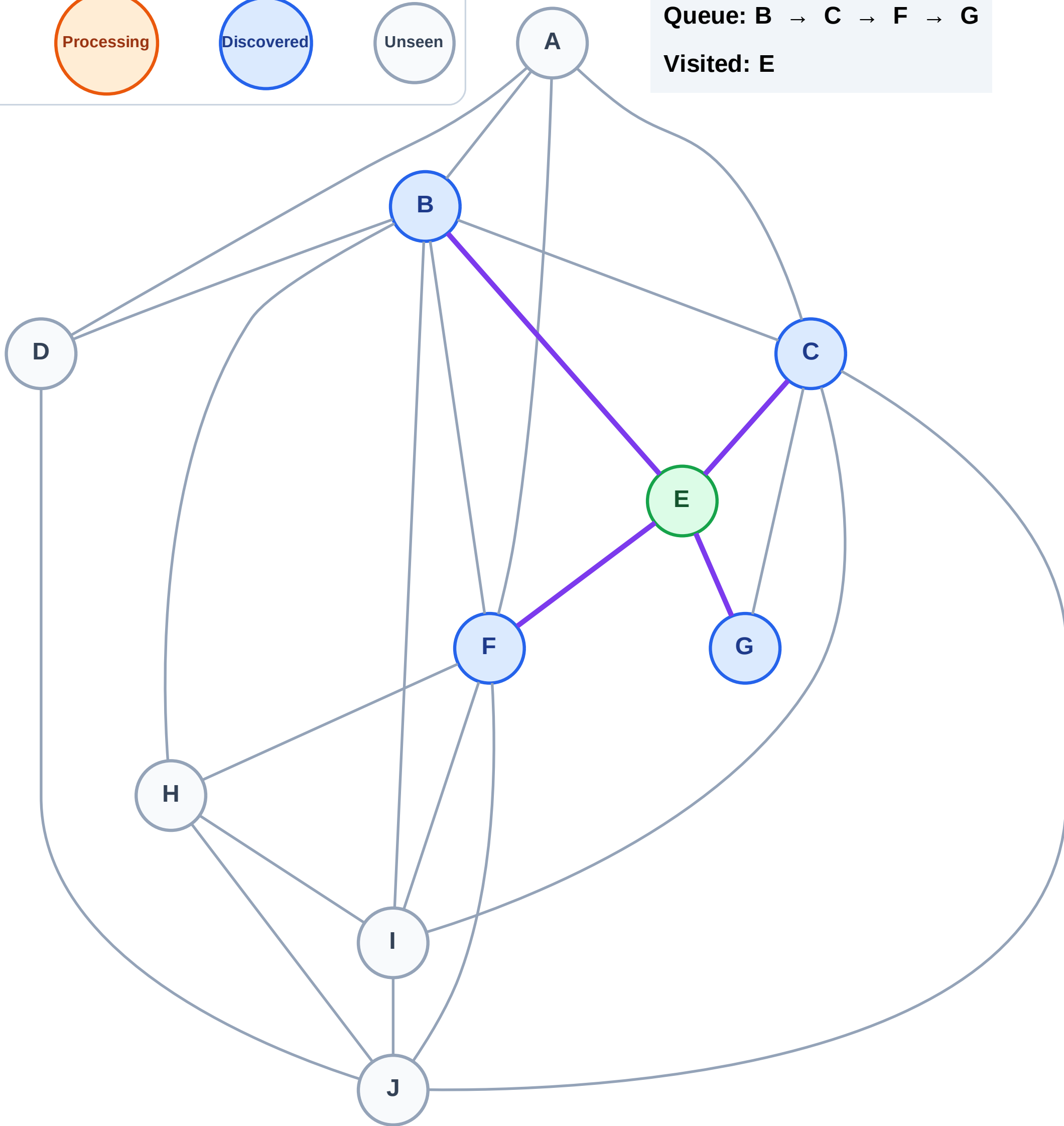
Step 3: Discover B, C, F, G from E

Legend



Queue: B → C → F → G

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

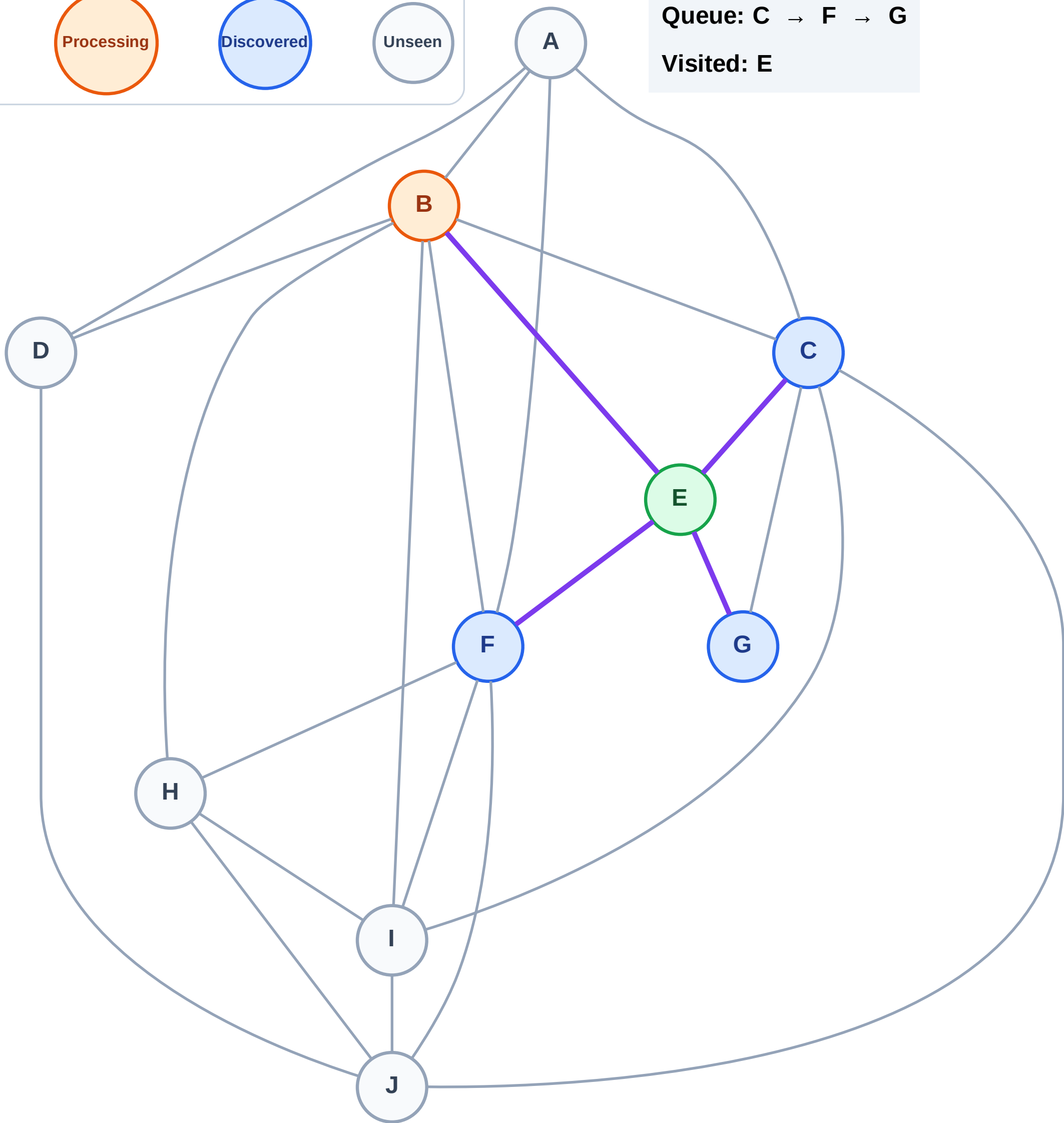
Processing

Discovered

Unseen

Queue: C → F → G

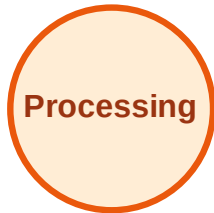
Visited: E



BFS Traversal

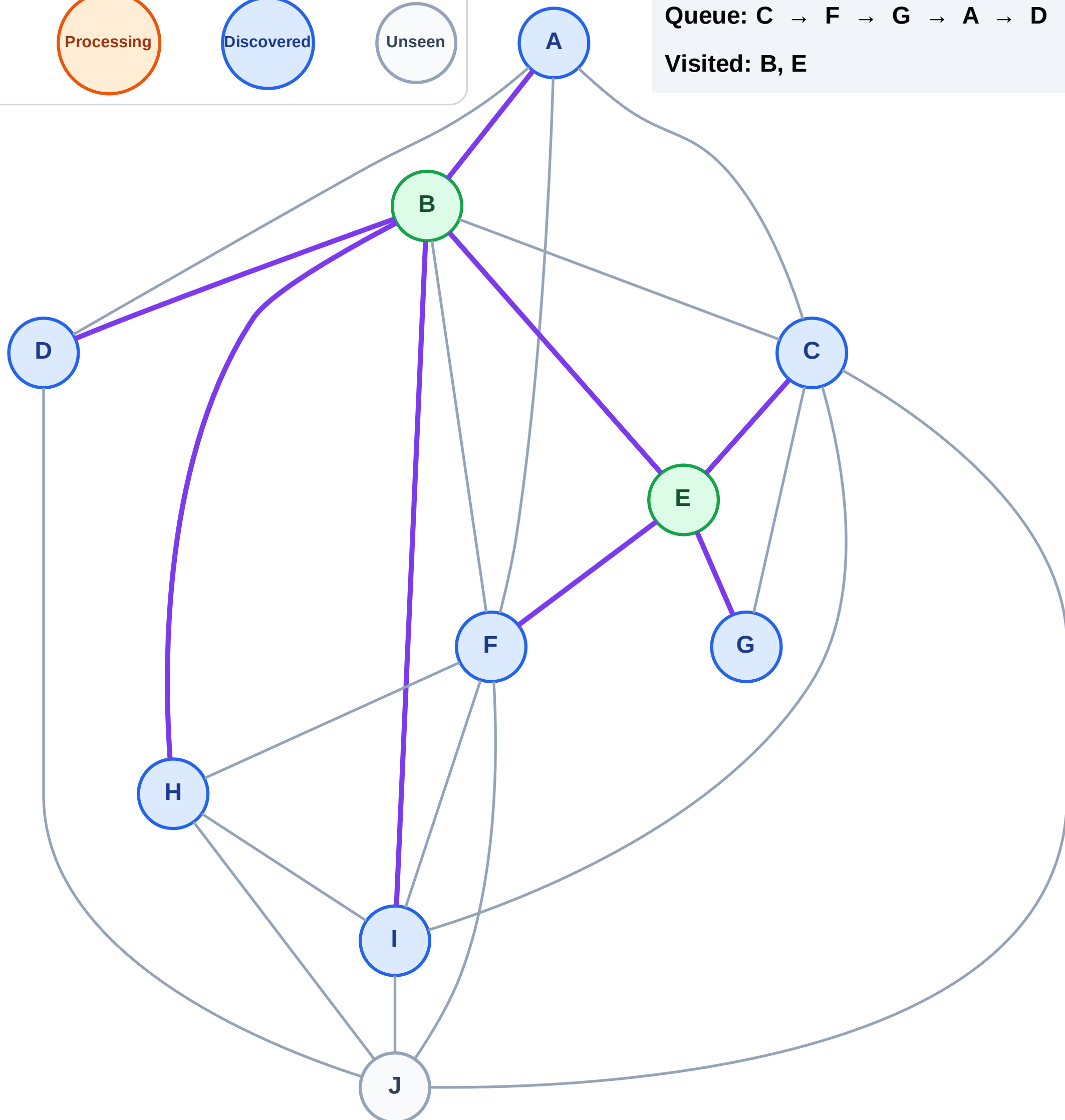
Step 5: Discover A, D, H, I from B

Legend



Queue: C → F → G → A → D → H → I

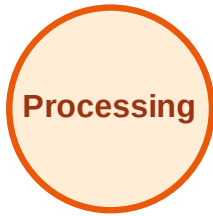
Visited: B, E



BFS Traversal

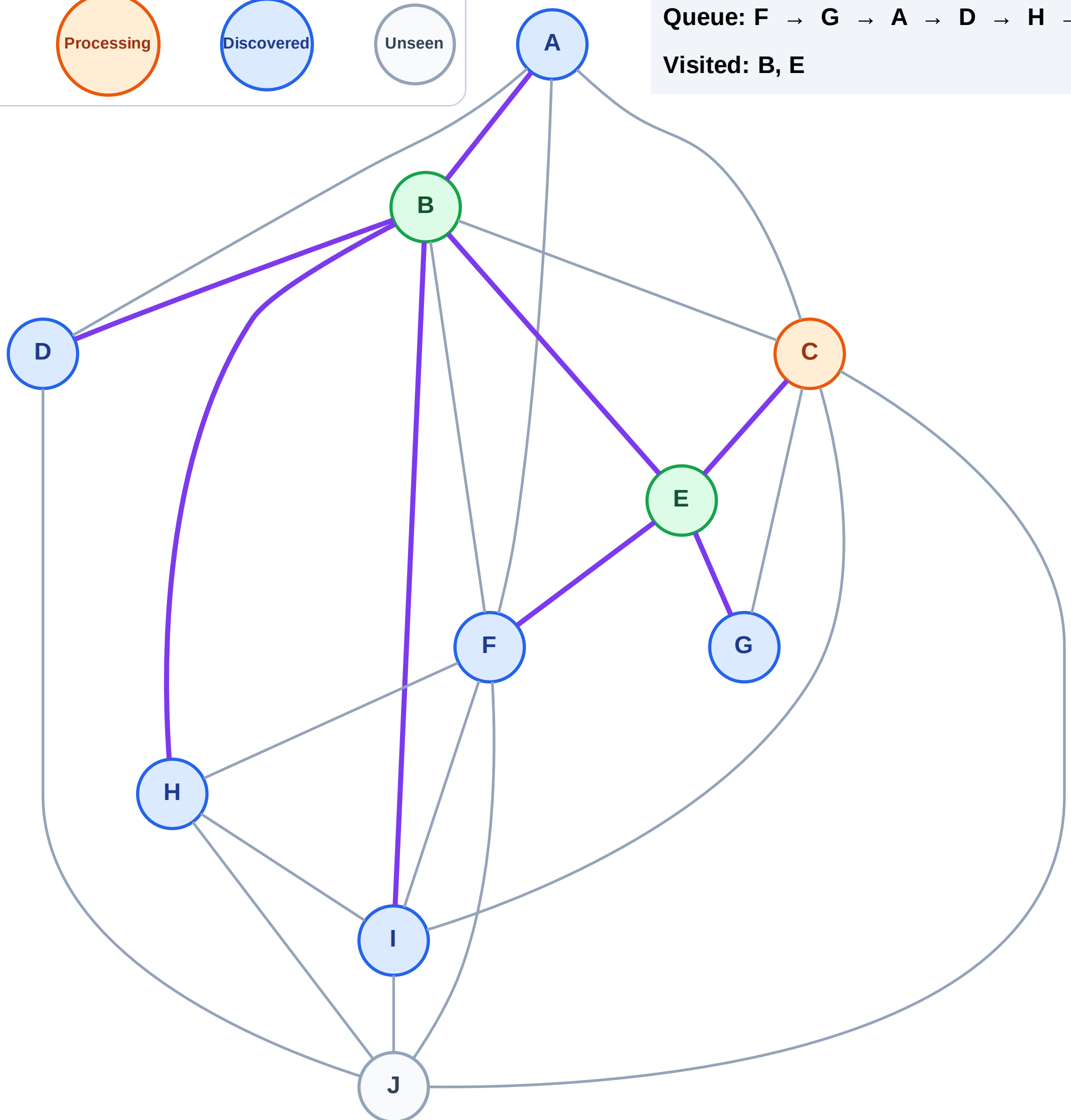
Step 6: Process node C

Legend



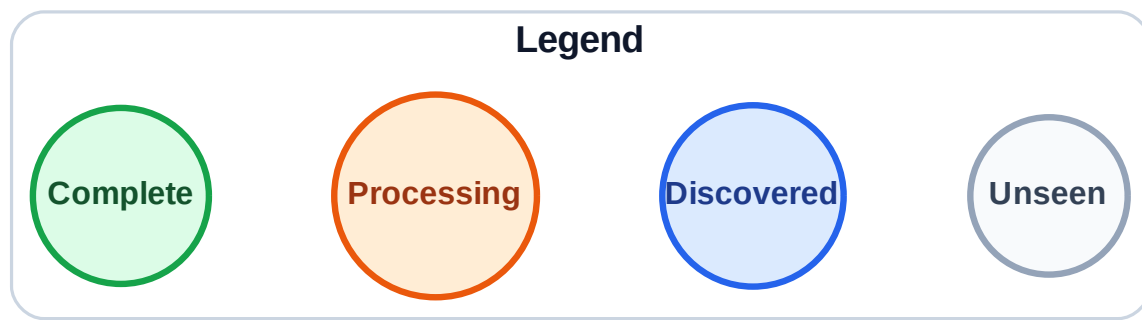
Queue: F → G → A → D → H → I

Visited: B, E



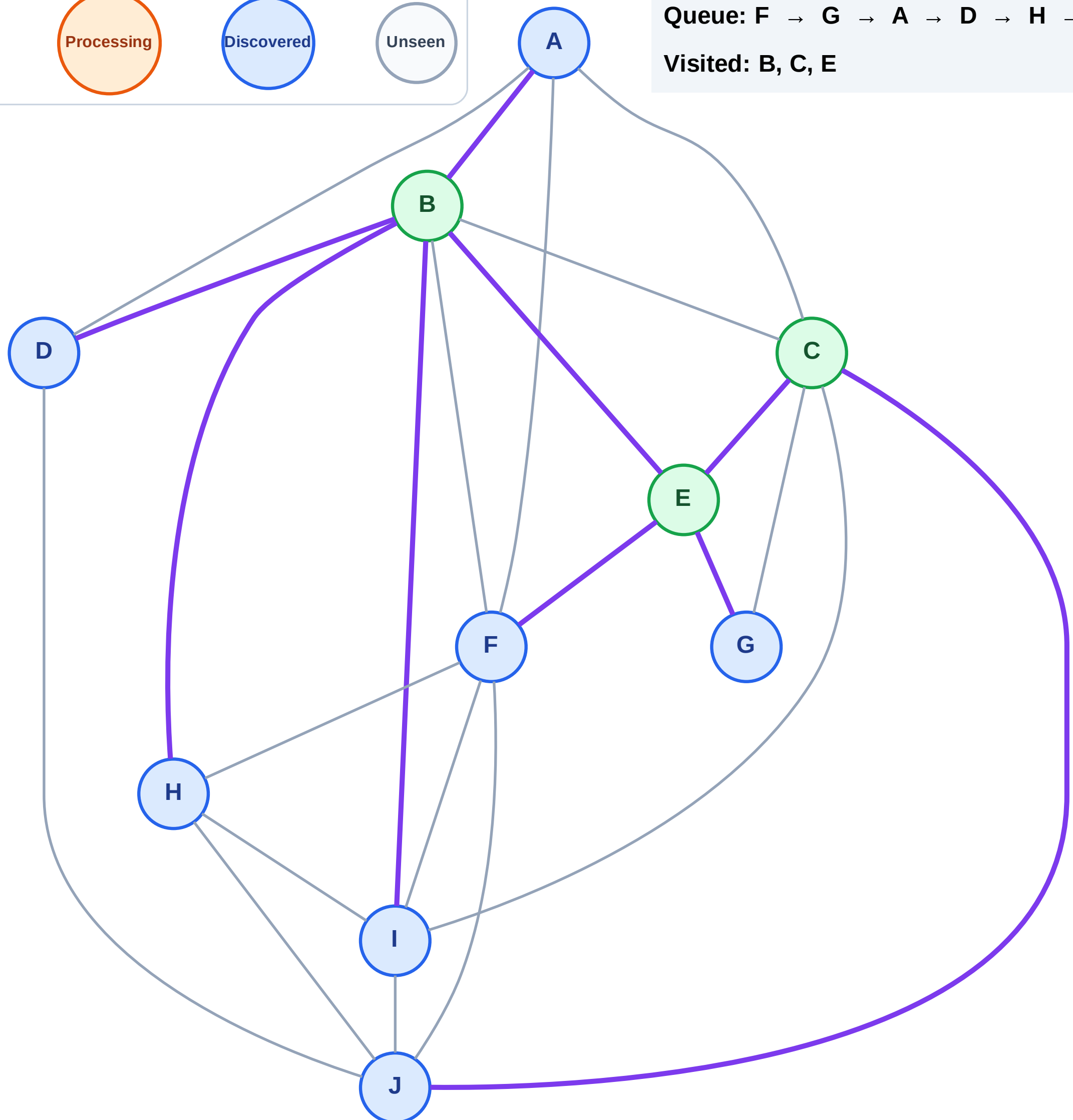
BFS Traversal

Step 7: Discover J from C



Queue: F → G → A → D → H → I → J

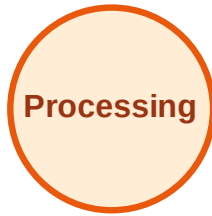
Visited: B, C, E



BFS Traversal

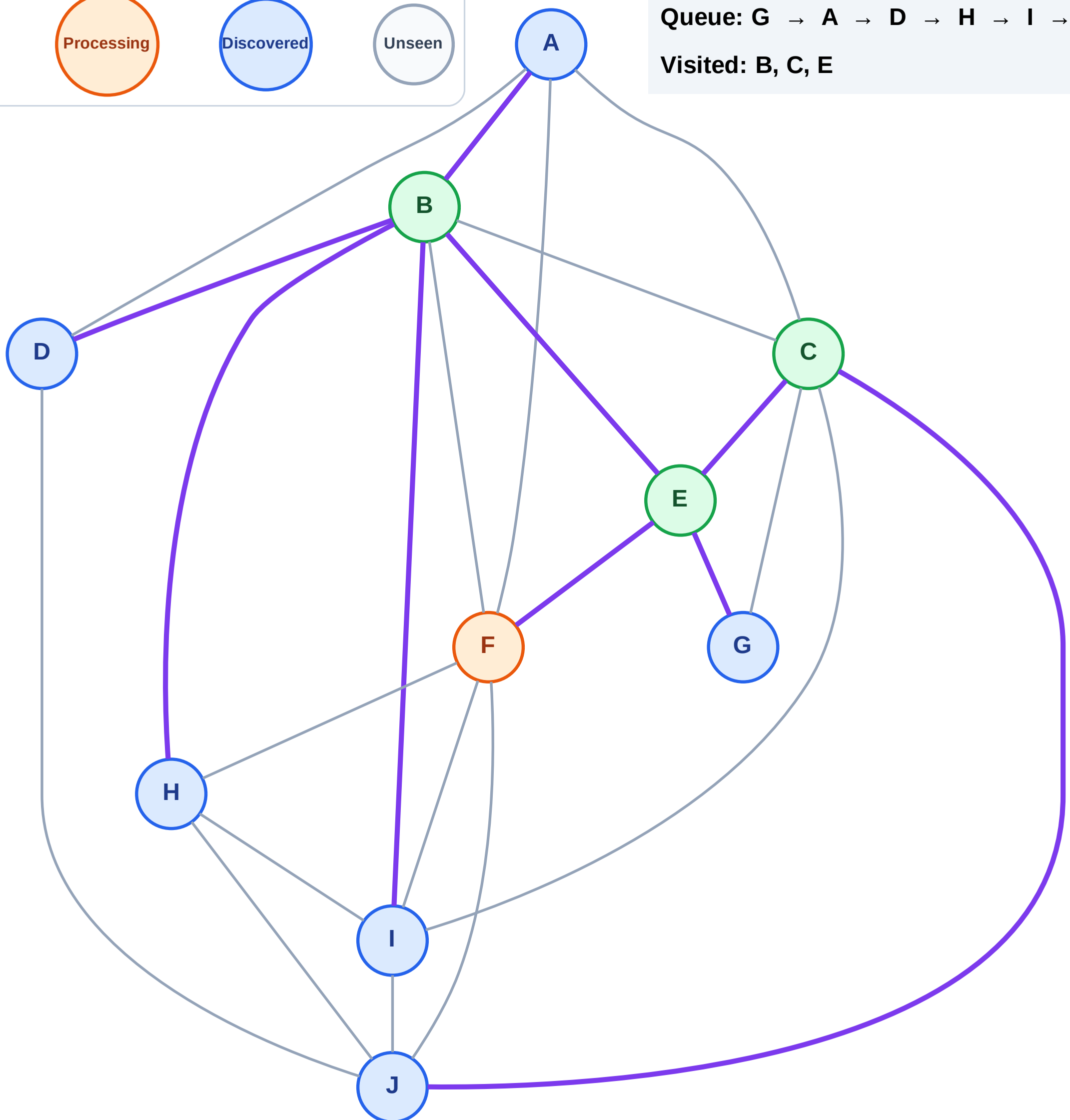
Step 8: Process node F

Legend



Queue: G → A → D → H → I → J

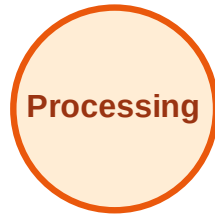
Visited: B, C, E



BFS Traversal

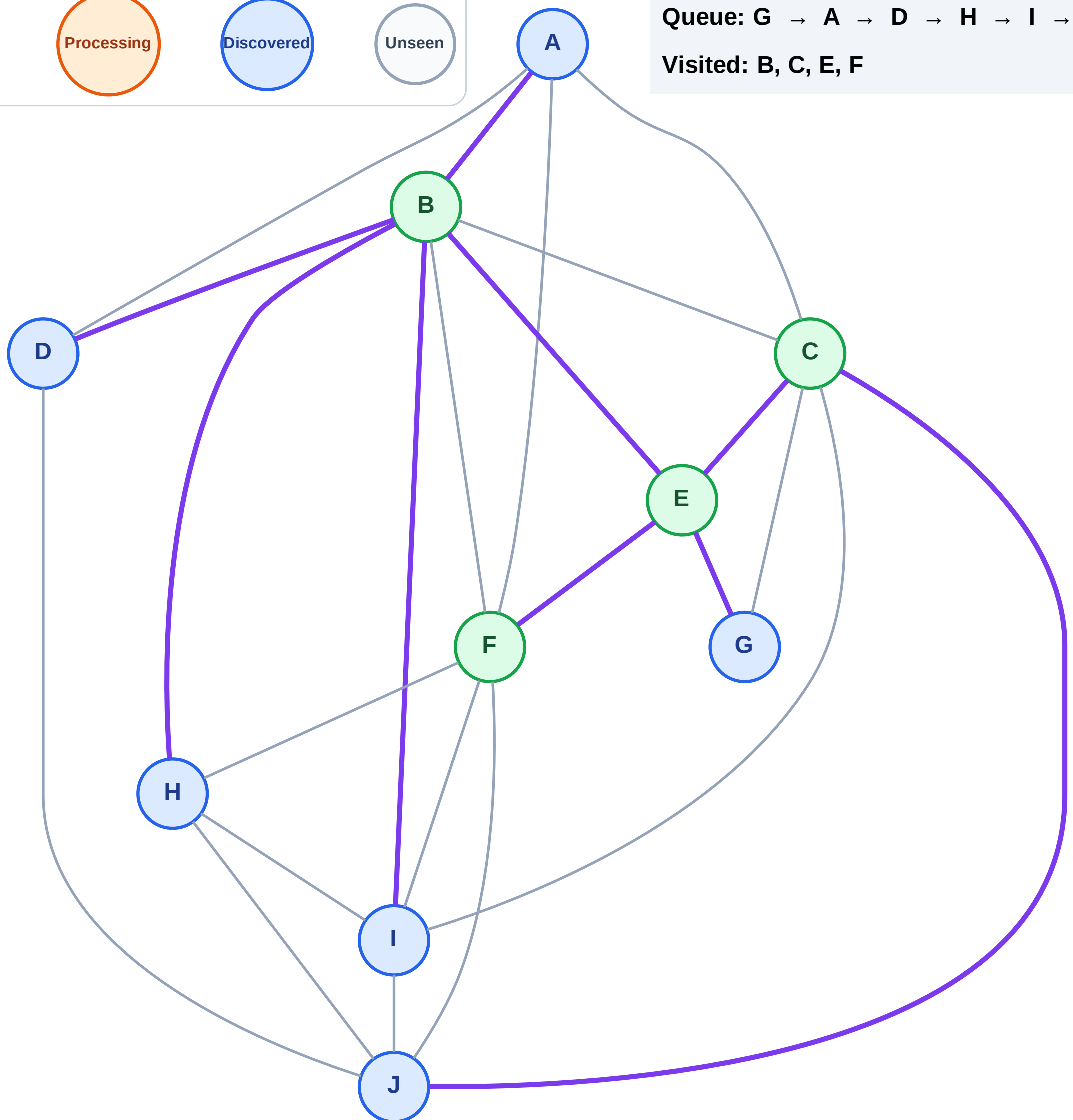
Step 9: No new neighbors from F

Legend



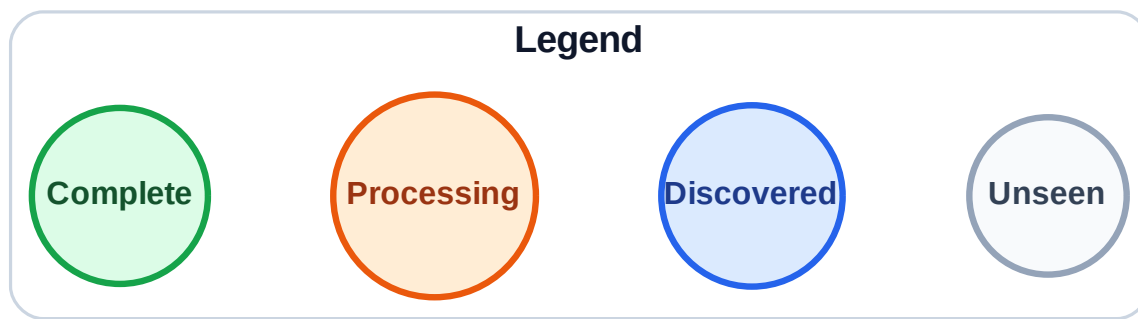
Queue: G → A → D → H → I → J

Visited: B, C, E, F

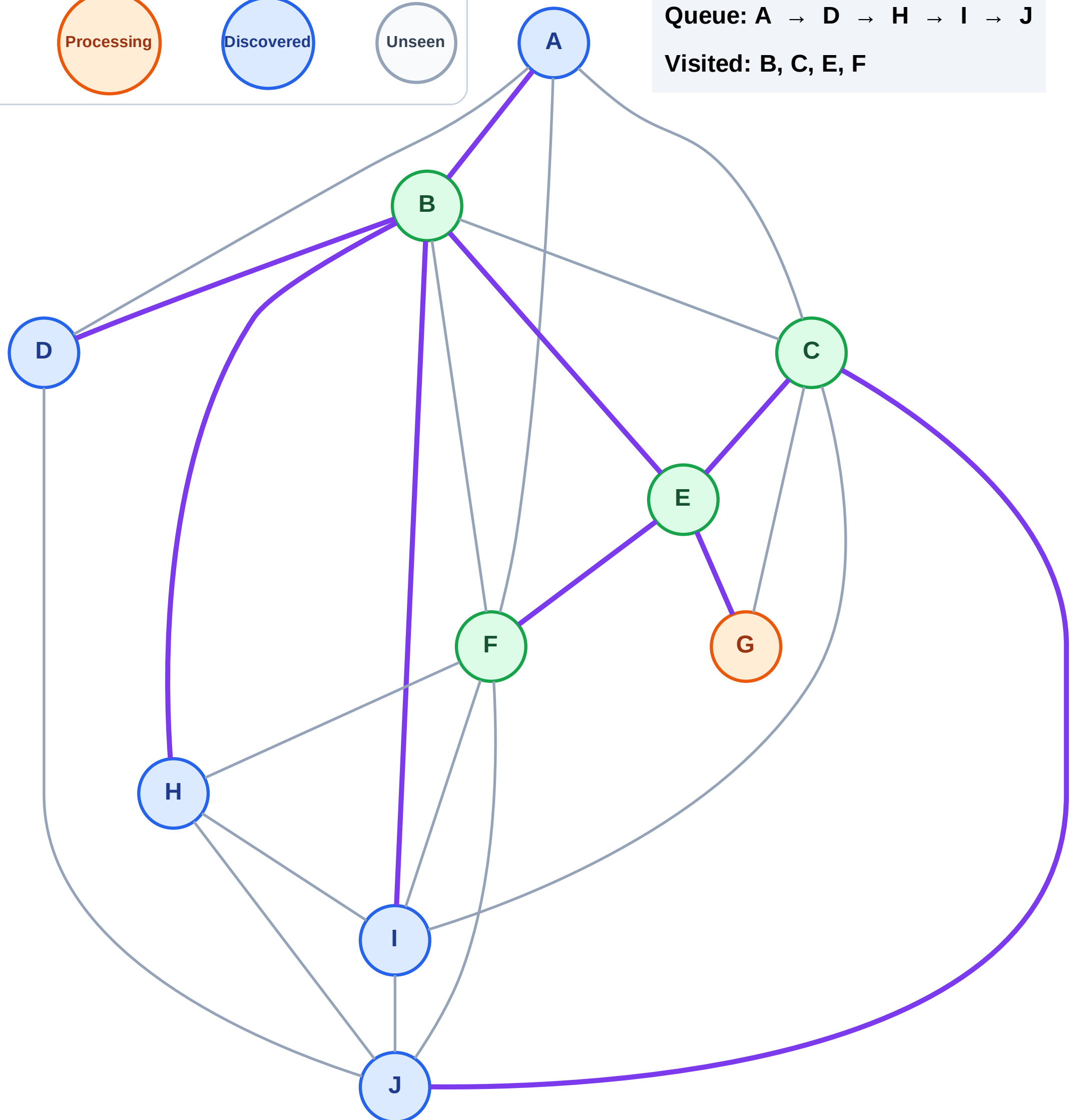


BFS Traversal

Step 10: Process node G



Queue: A → D → H → I → J
Visited: B, C, E, F



BFS Traversal

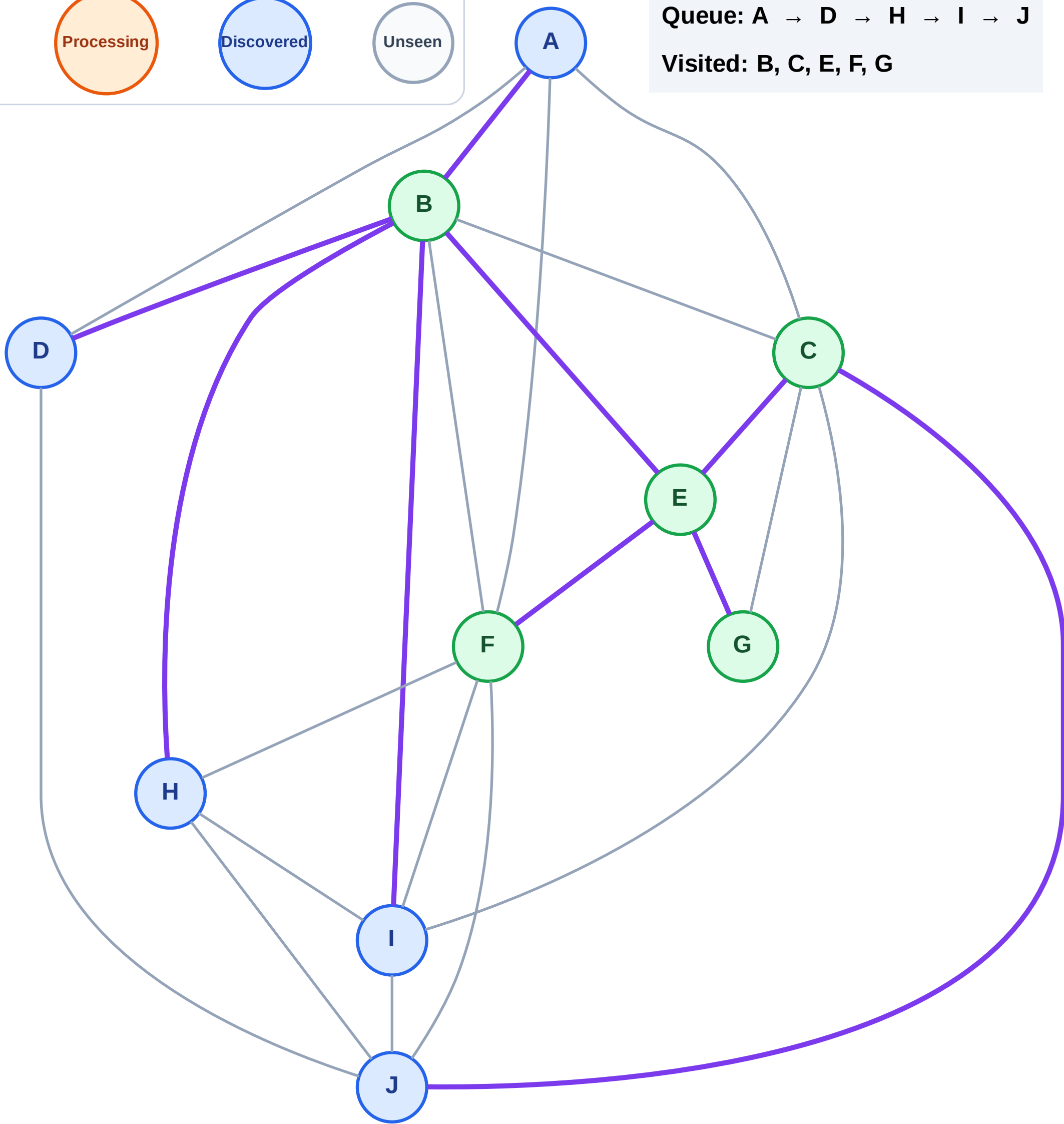
Step 11: No new neighbors from G

Legend



Queue: A → D → H → I → J

Visited: B, C, E, F, G



BFS Traversal

Step 12: Process node A

Legend

Complete

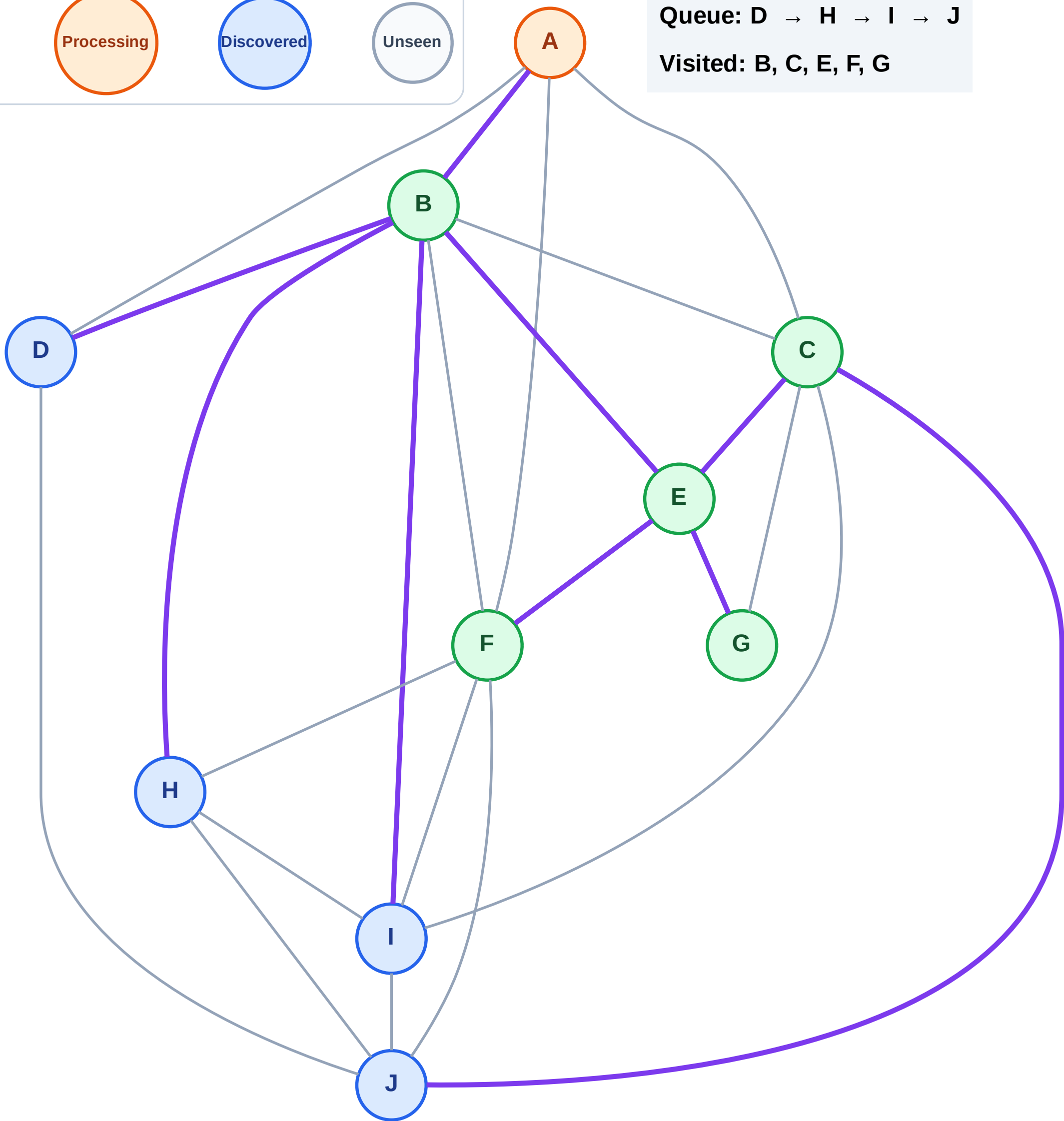
Processing

Discovered

Unseen

Queue: D → H → I → J

Visited: B, C, E, F, G



BFS Traversal

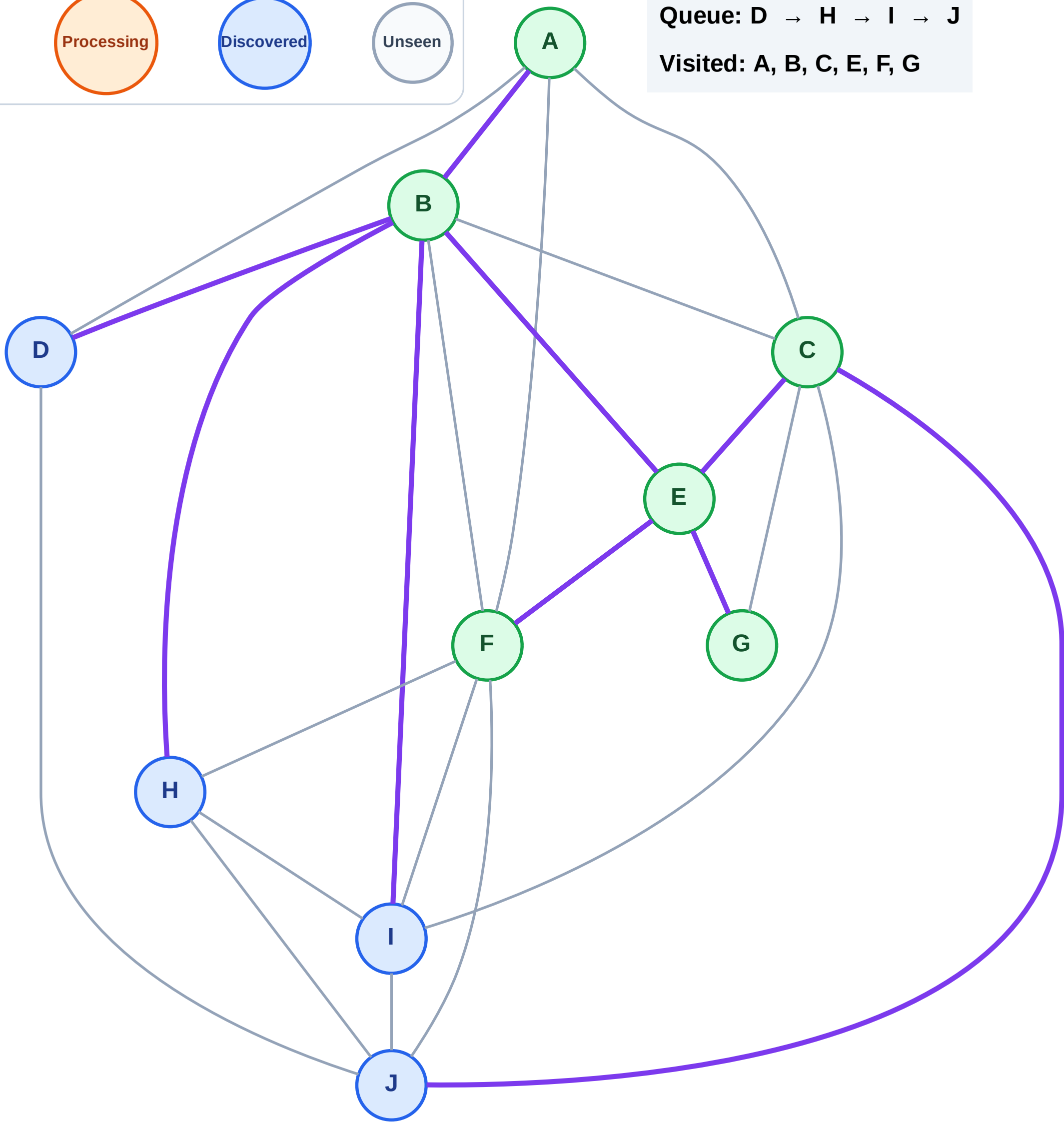
Step 13: No new neighbors from A

Legend



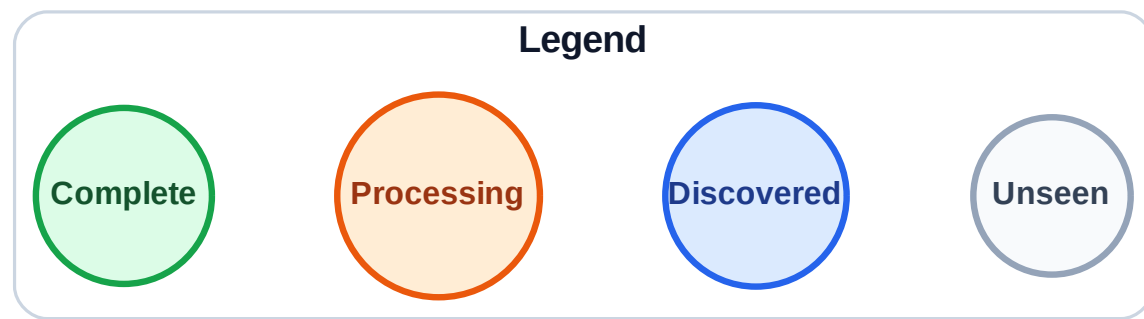
Queue: D → H → I → J

Visited: A, B, C, E, F, G

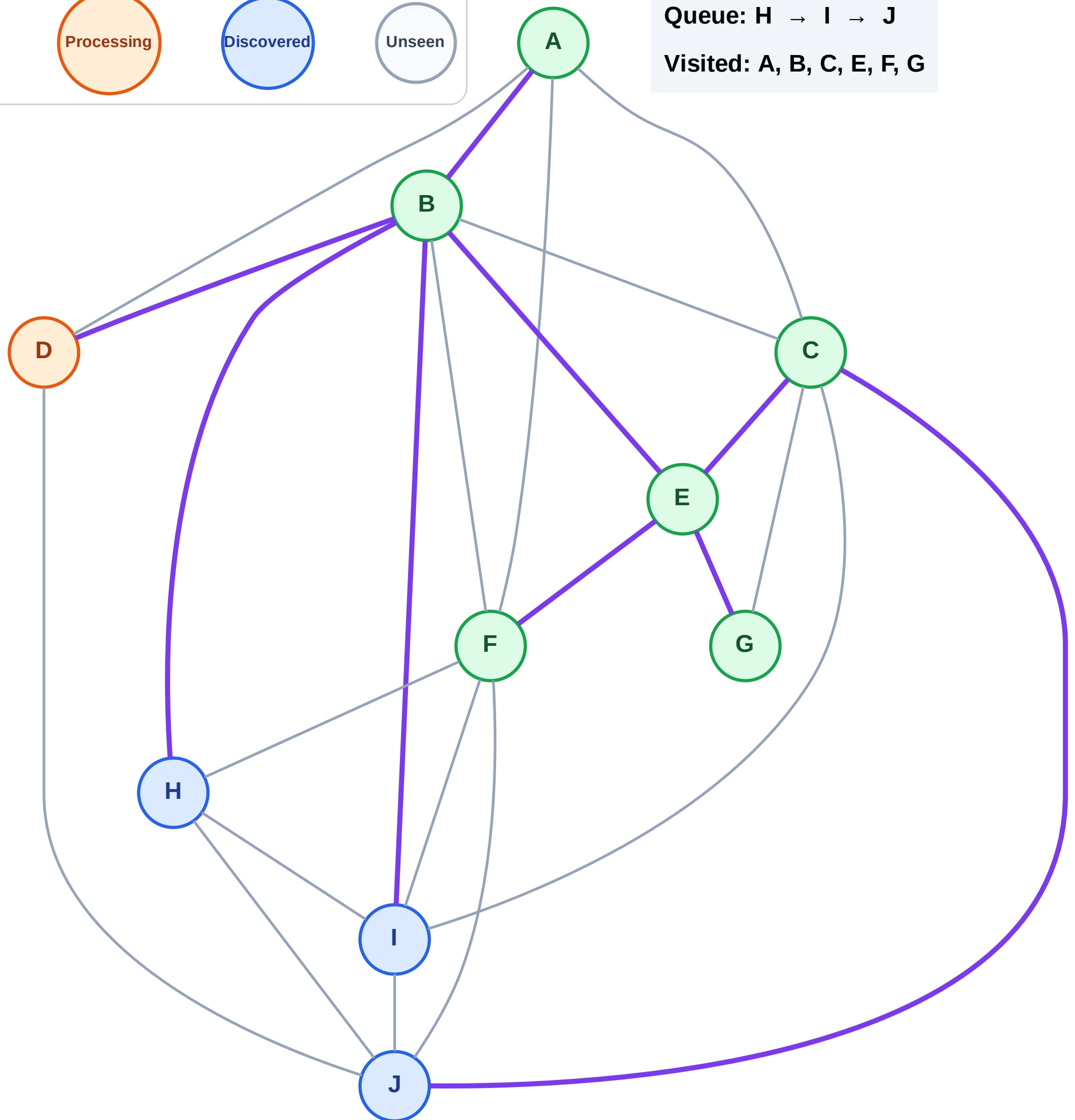


BFS Traversal

Step 14: Process node D



Queue: H → I → J
Visited: A, B, C, E, F, G



BFS Traversal

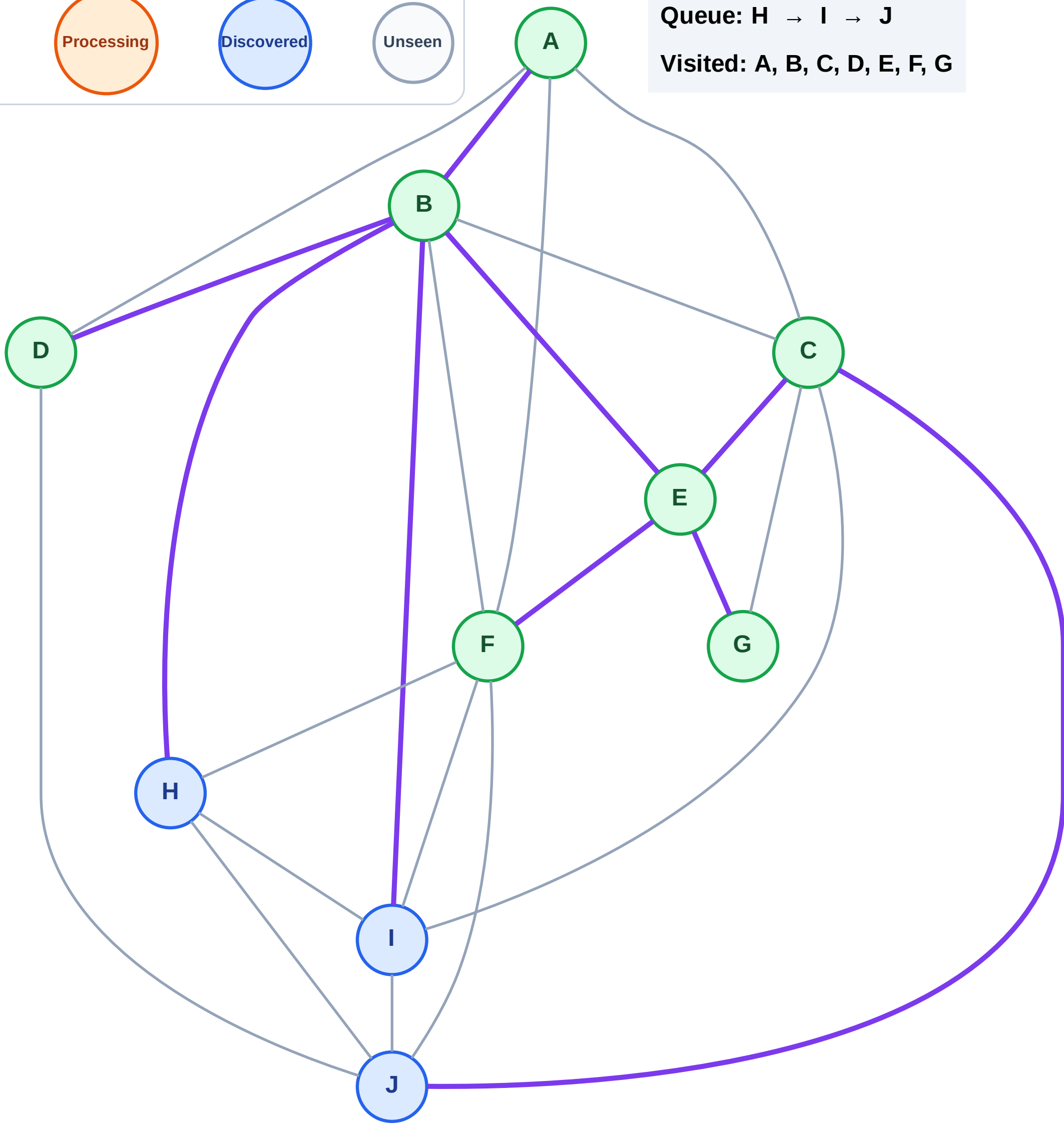
Step 15: No new neighbors from D

Legend



Queue: H → I → J

Visited: A, B, C, D, E, F, G



BFS Traversal

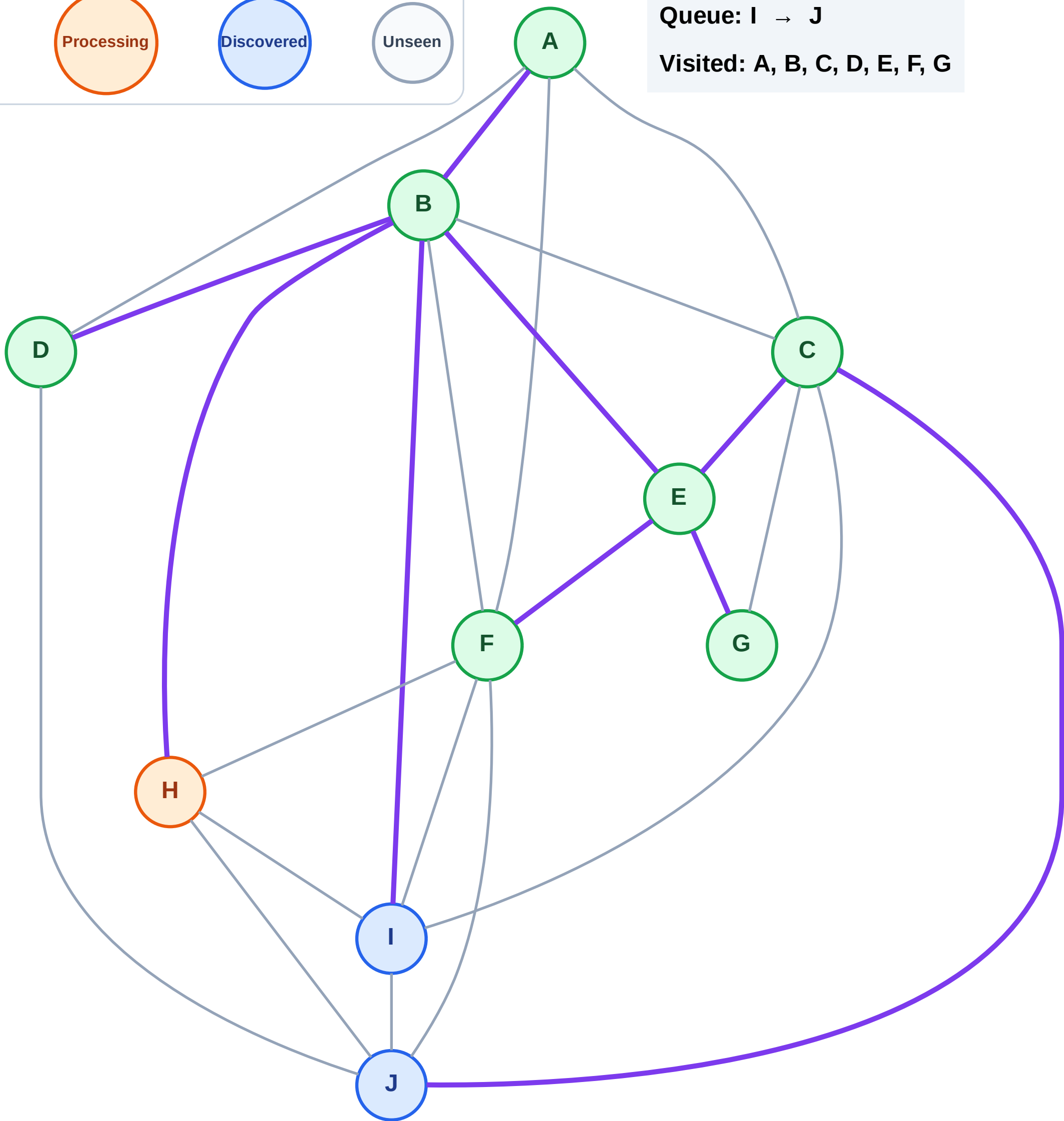
Step 16: Process node H

Legend



Queue: I → J

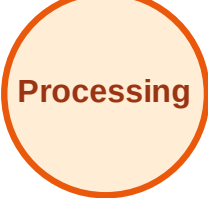
Visited: A, B, C, D, E, F, G



BFS Traversal

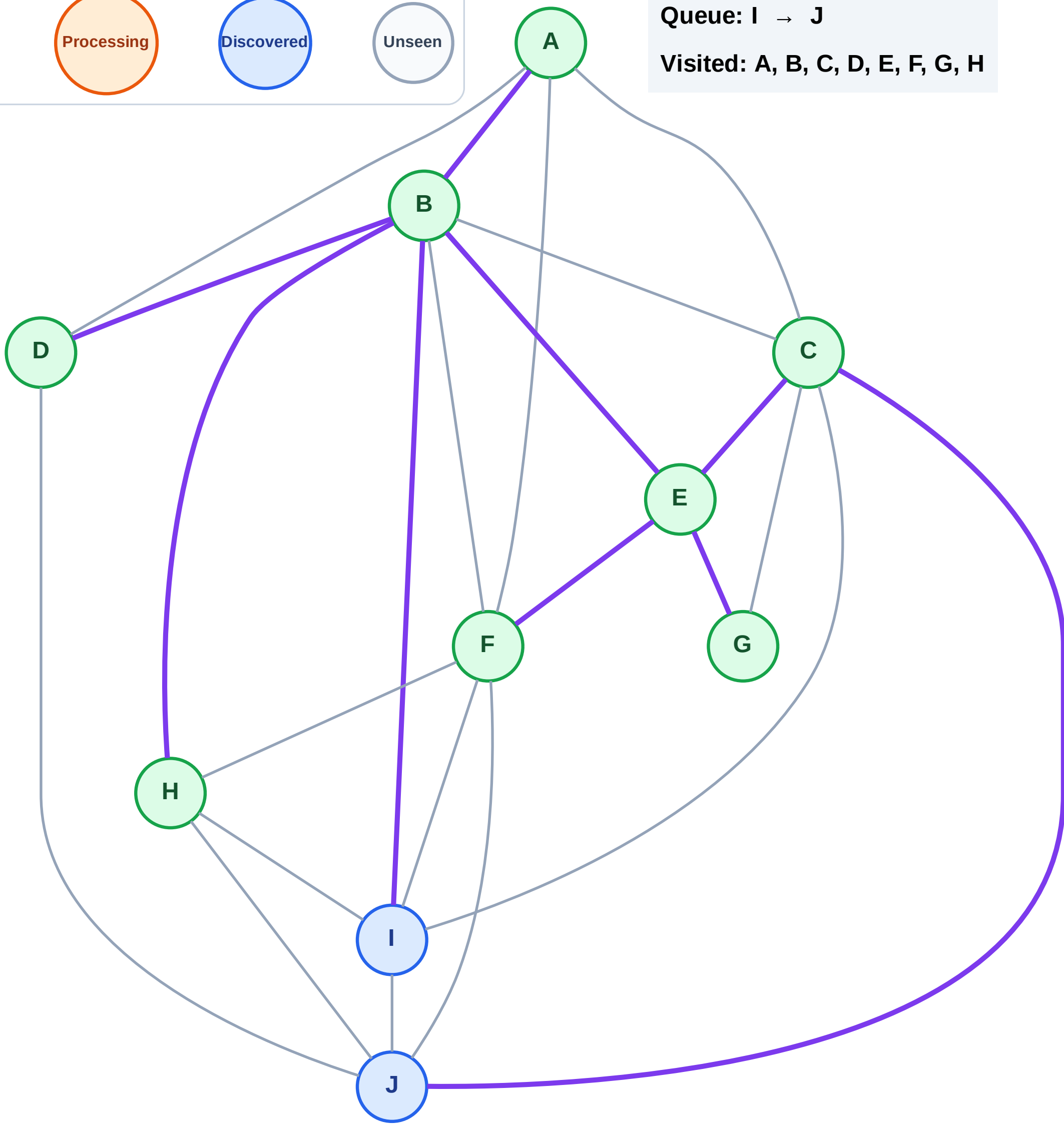
Step 17: No new neighbors from H

Legend



Queue: I → J

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node I

Legend

Complete

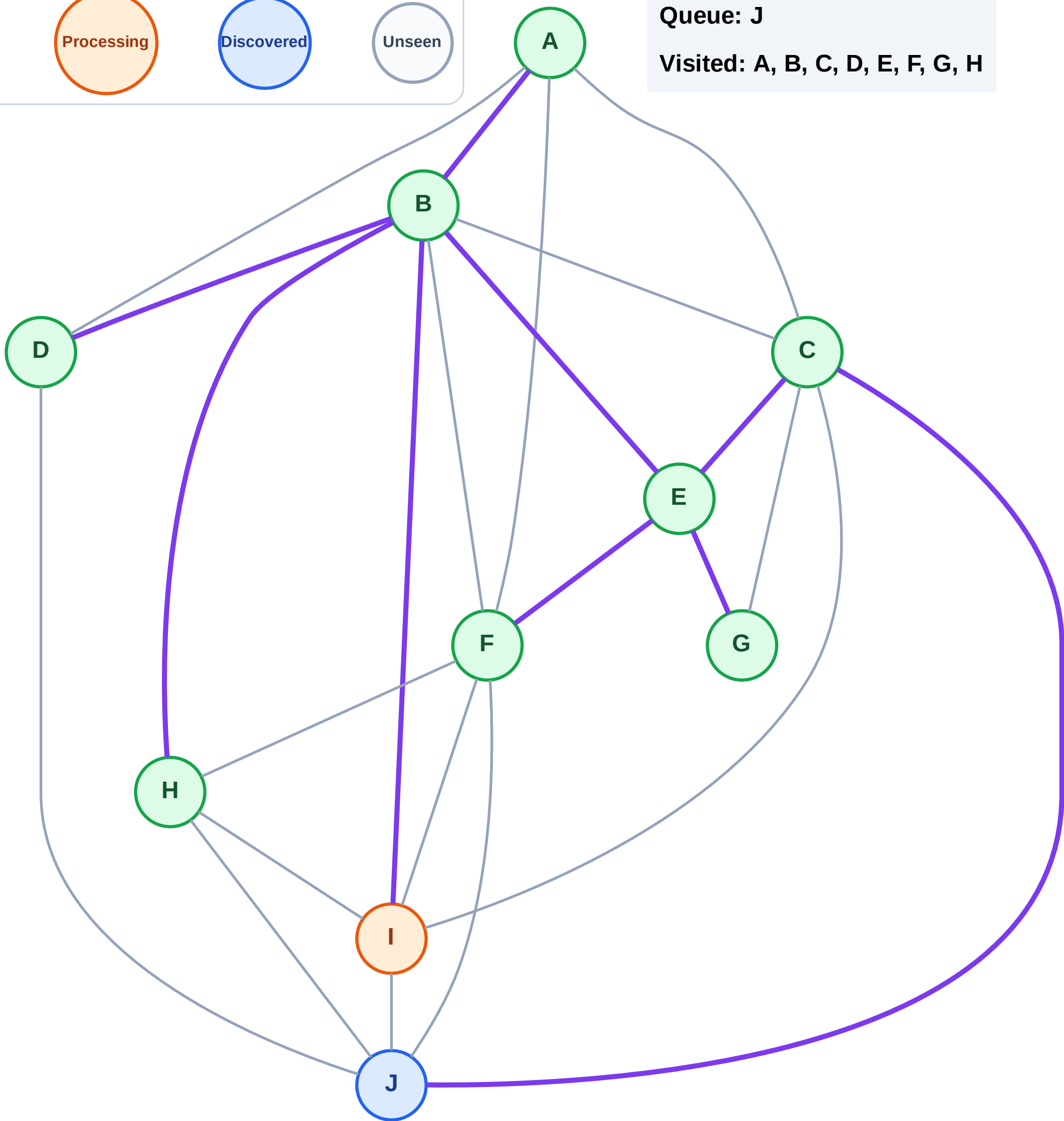
Processing

Discovered

Unseen

Queue: J

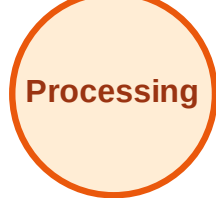
Visited: A, B, C, D, E, F, G, H



BFS Traversal

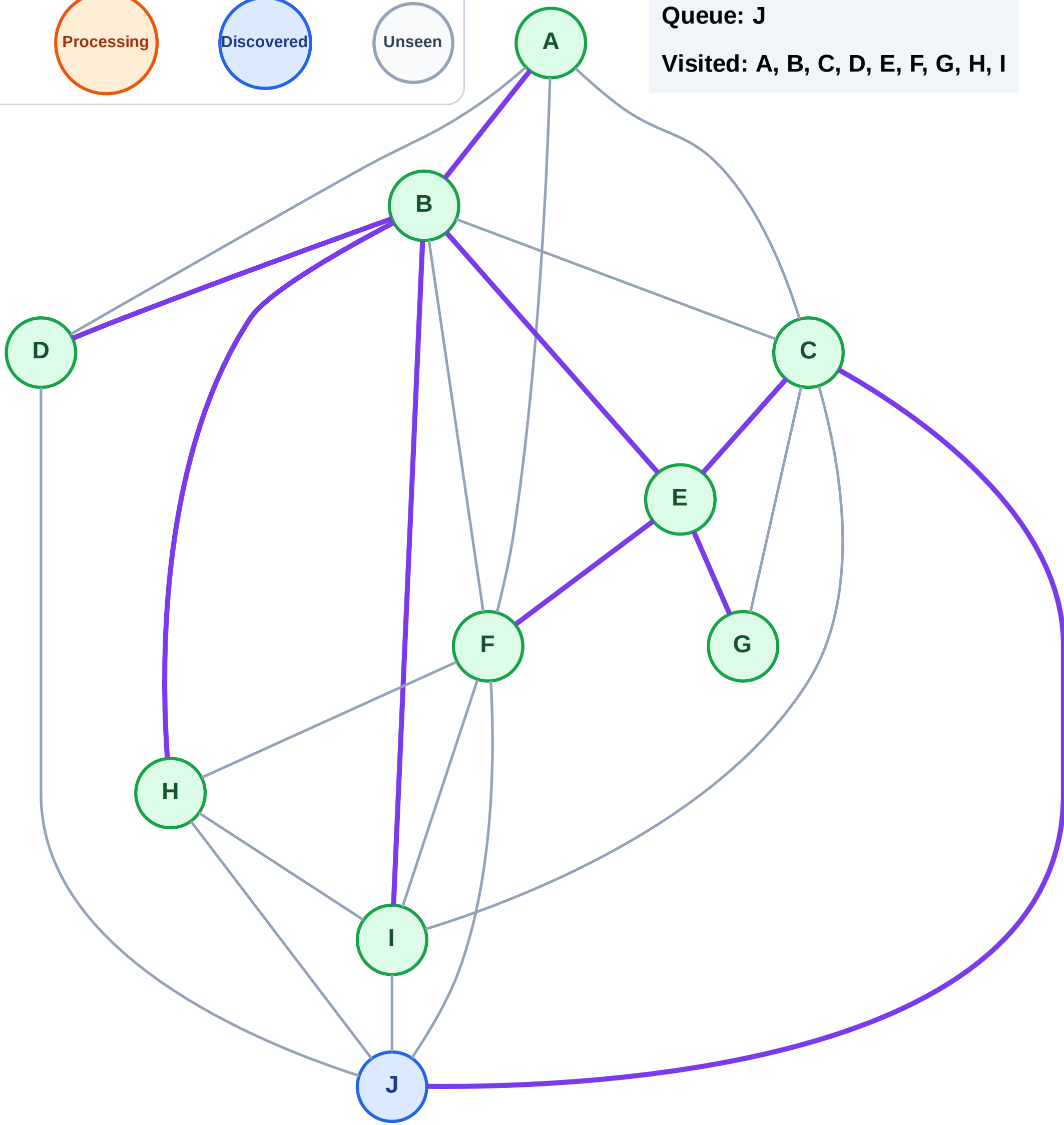
Step 19: No new neighbors from I

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

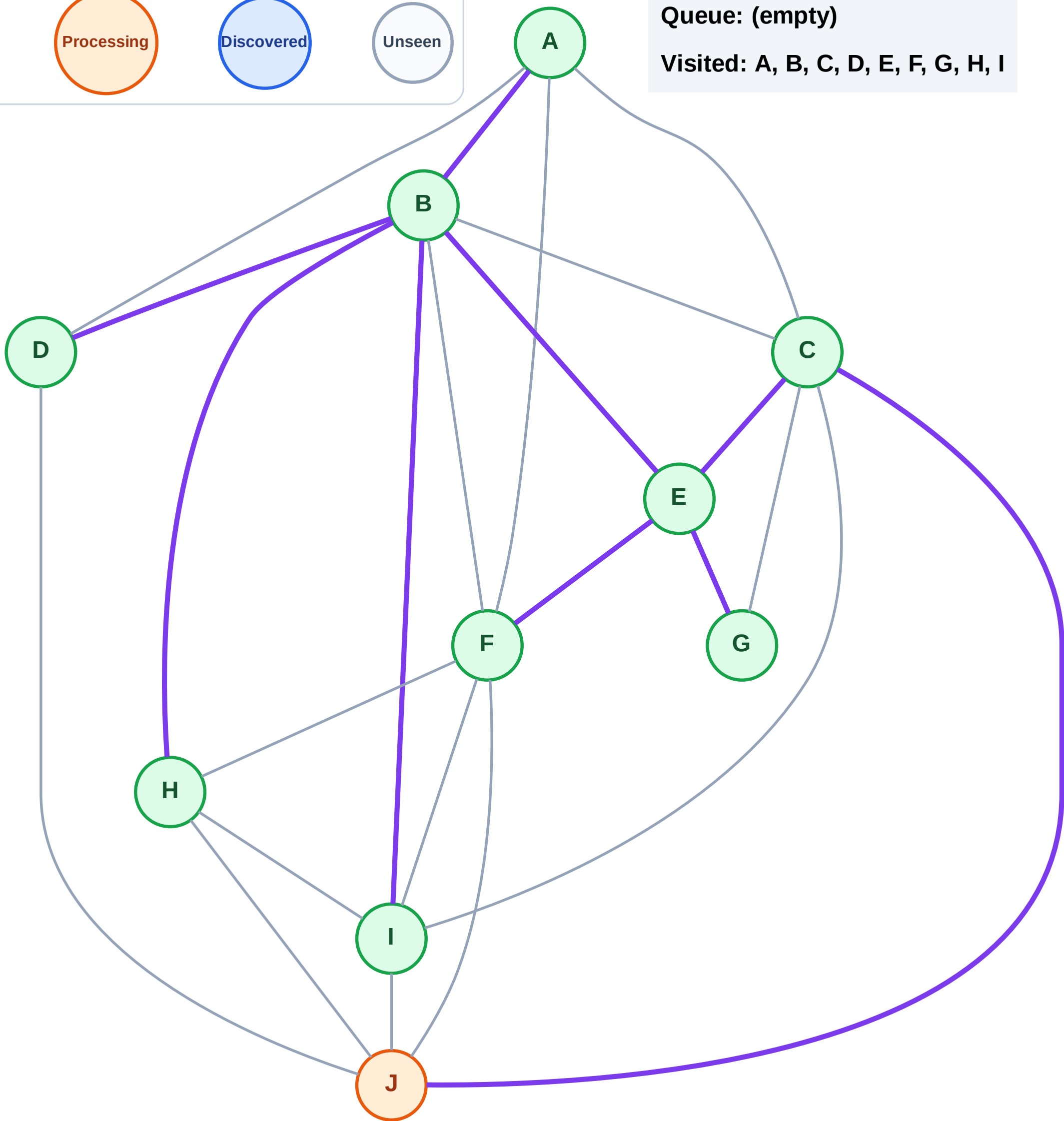
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

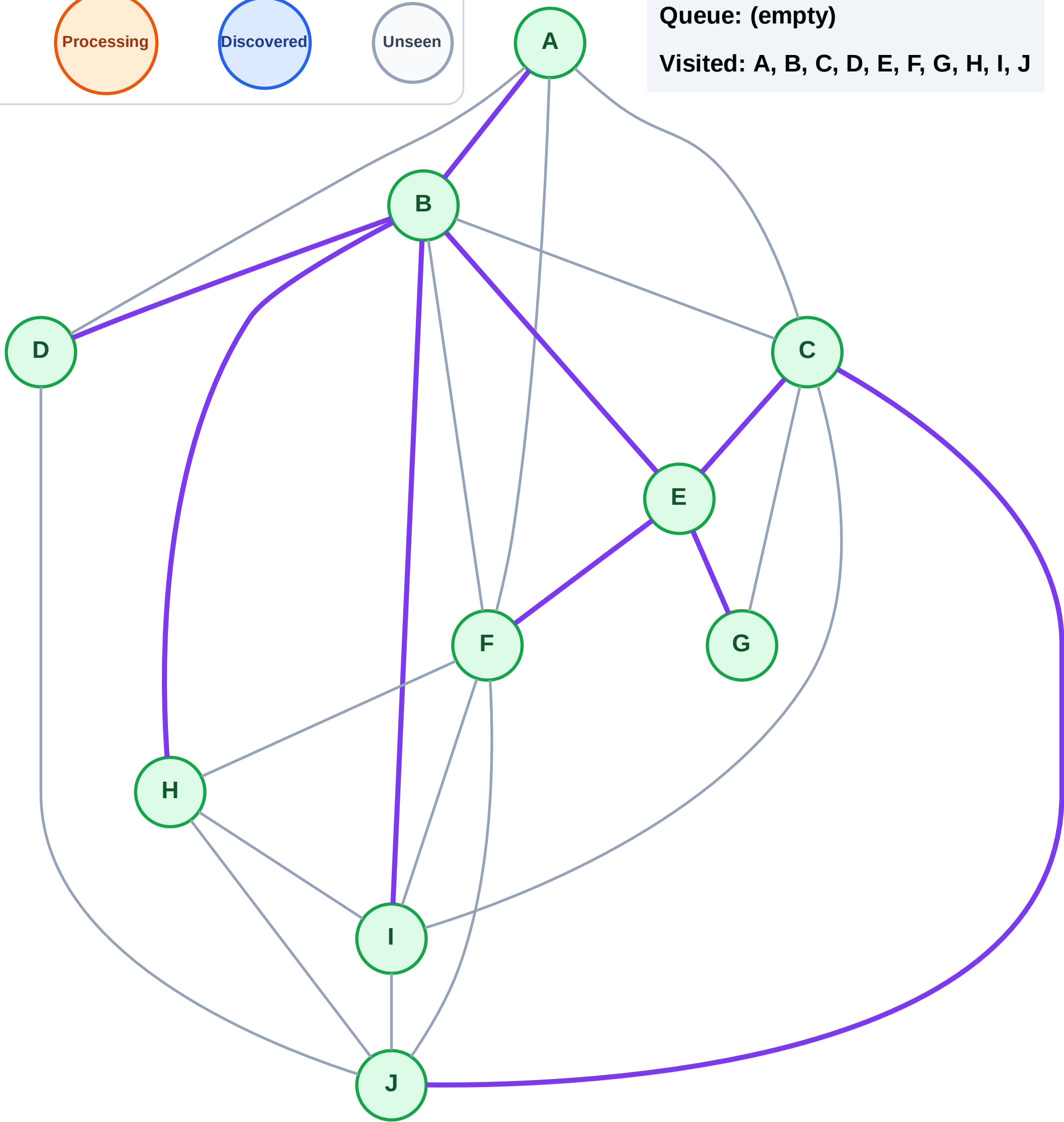
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

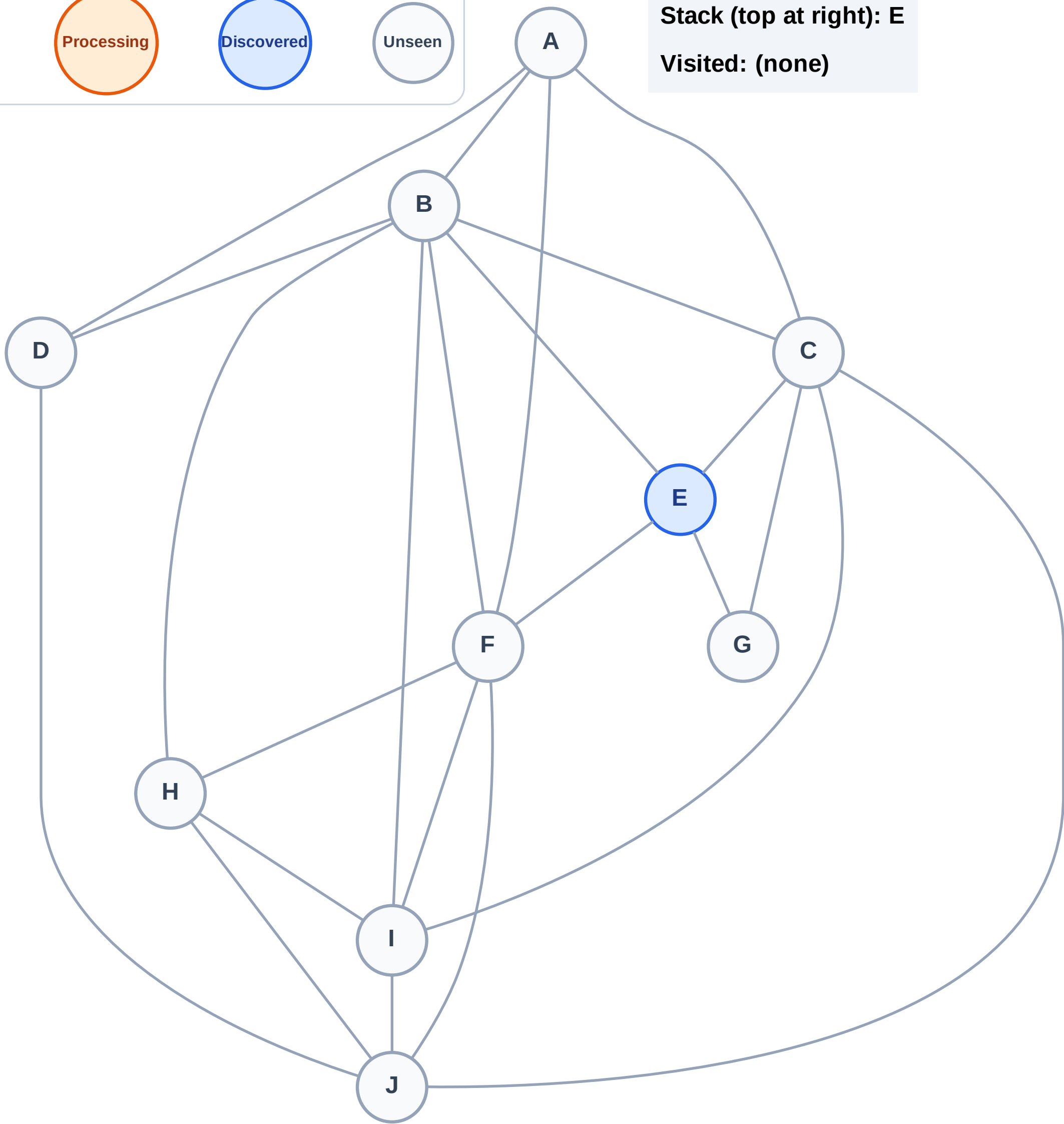
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

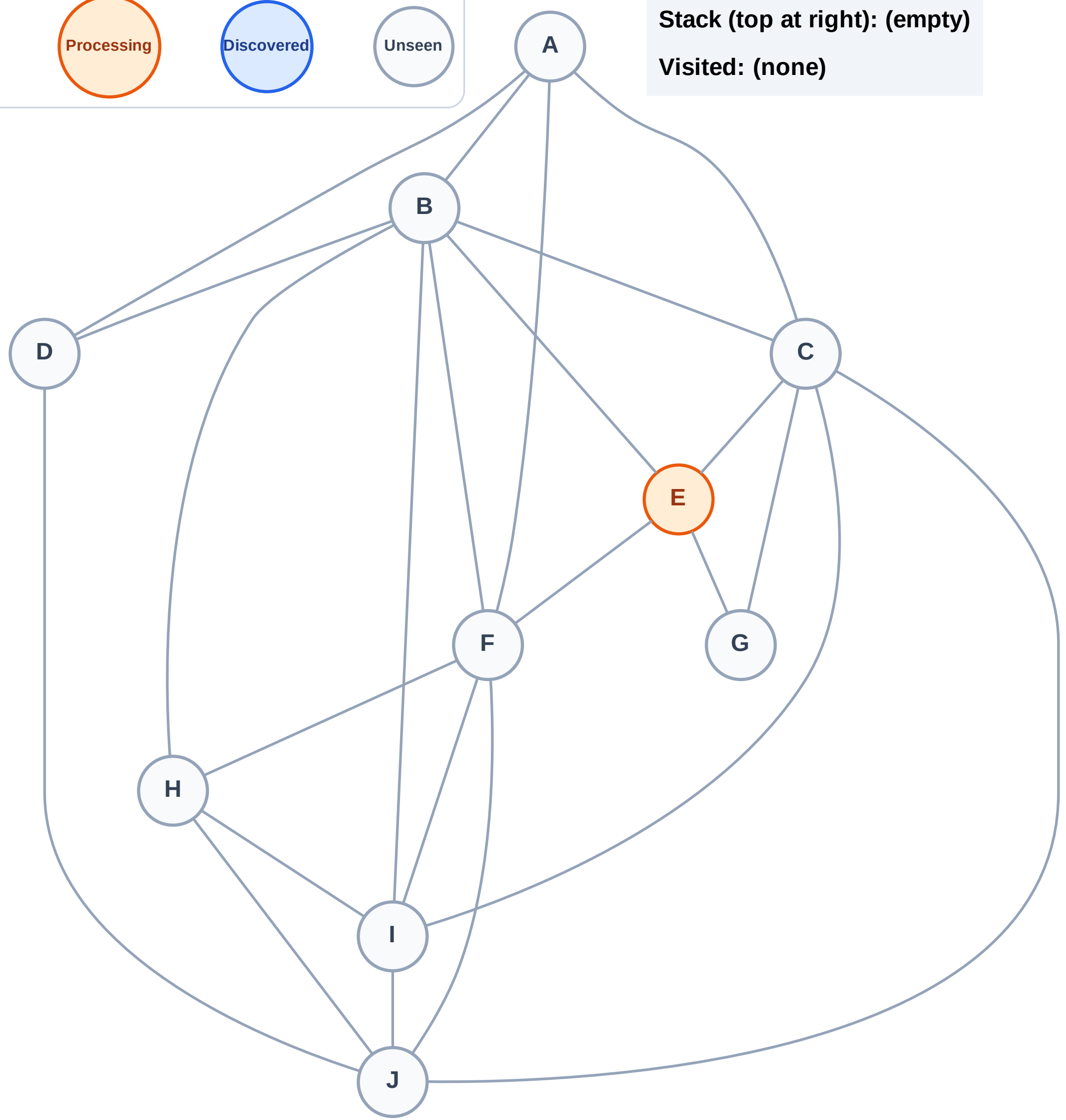
Step 2: Process node E

Legend



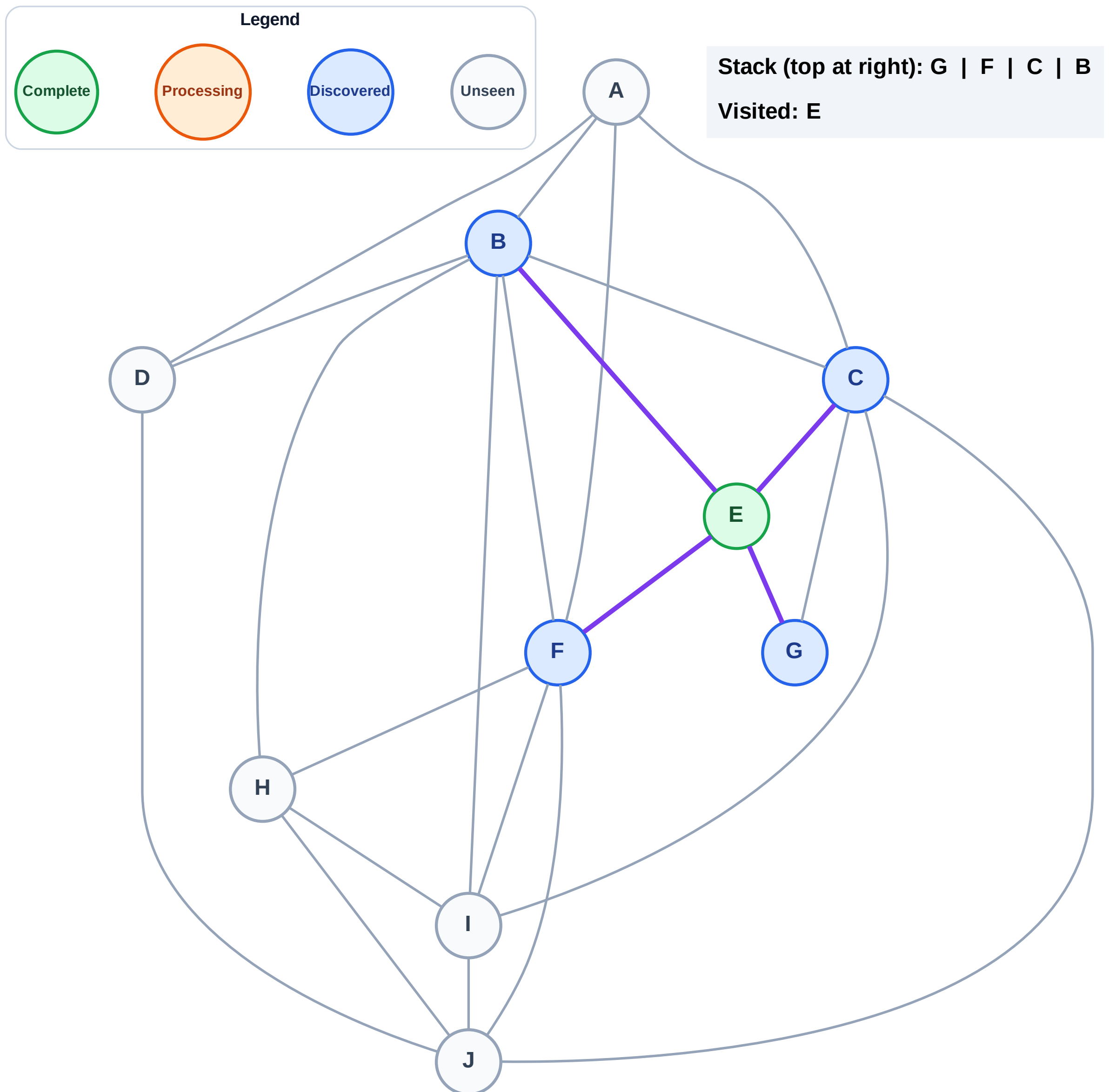
Stack (top at right): (empty)

Visited: (none)



Step 3: Discover G, F, C, B from E

Step 3: Discover G, F, C, B from E



DFS Traversal

Step 4: Process node B

Legend

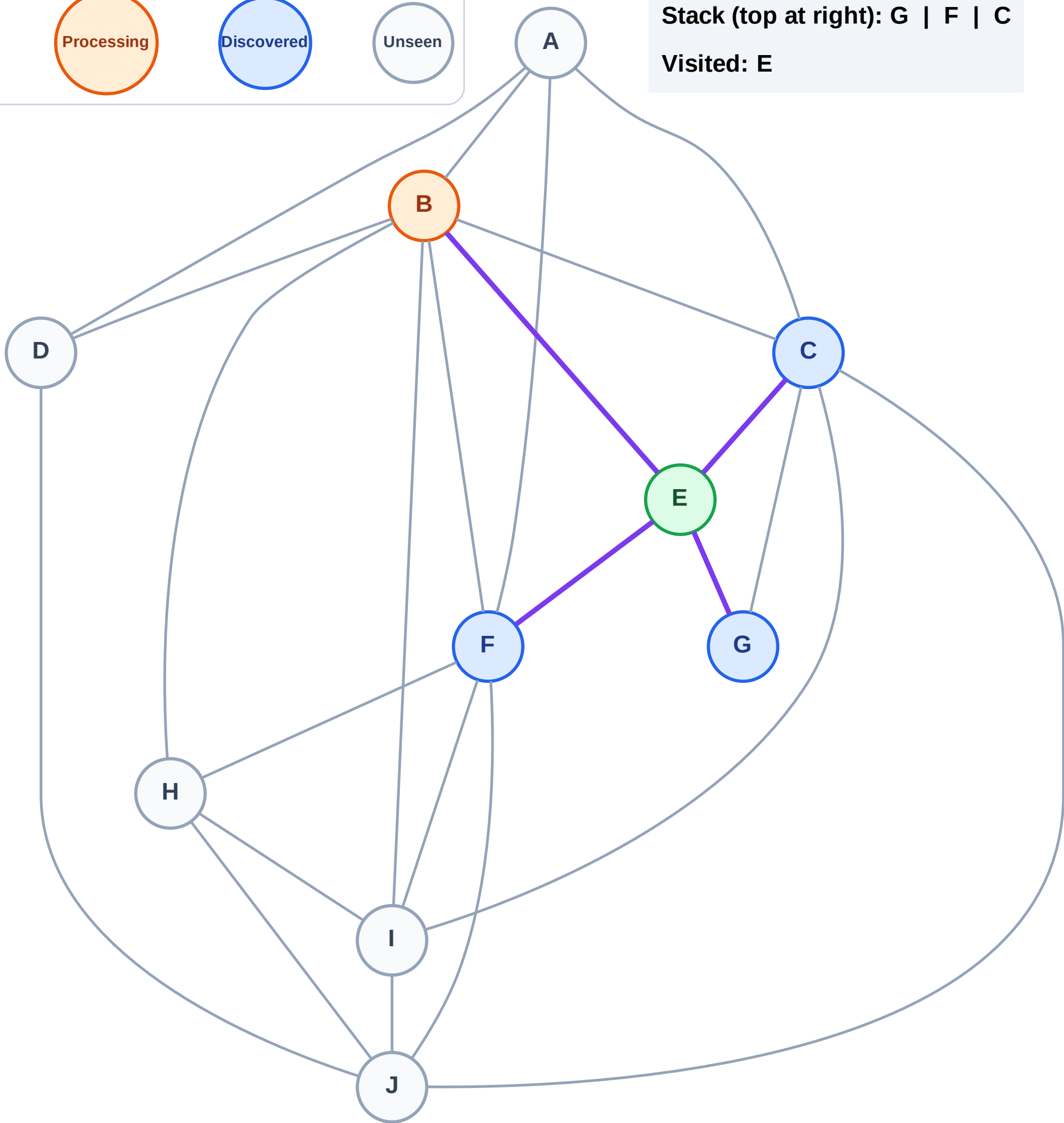
Complete

Processing

Discovered

Unseen

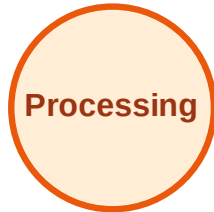
Stack (top at right): G | F | C
Visited: E



DFS Traversal

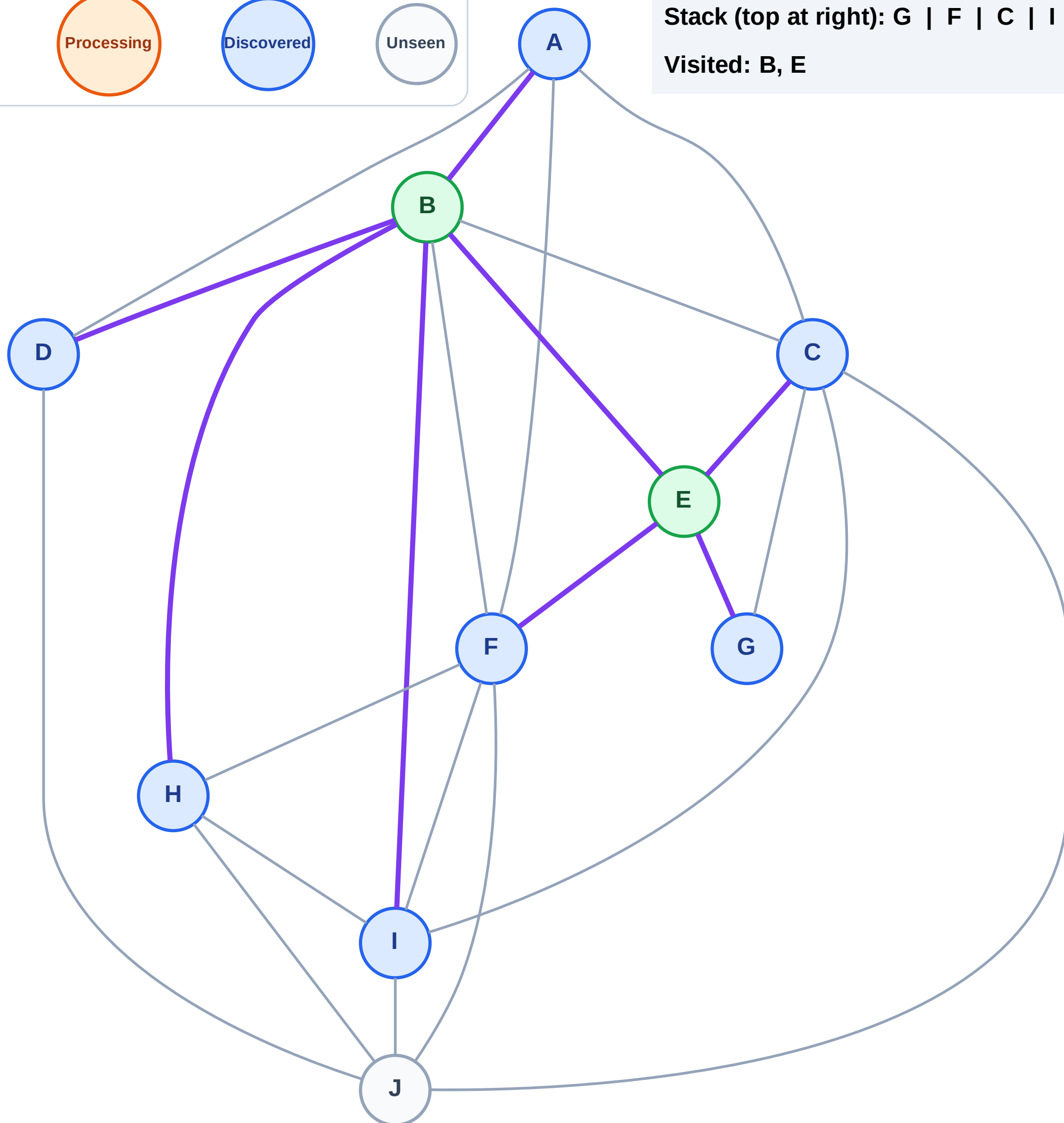
Step 5: Discover I, H, D, A from B

Legend



Stack (top at right): G | F | C | I | H | D | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

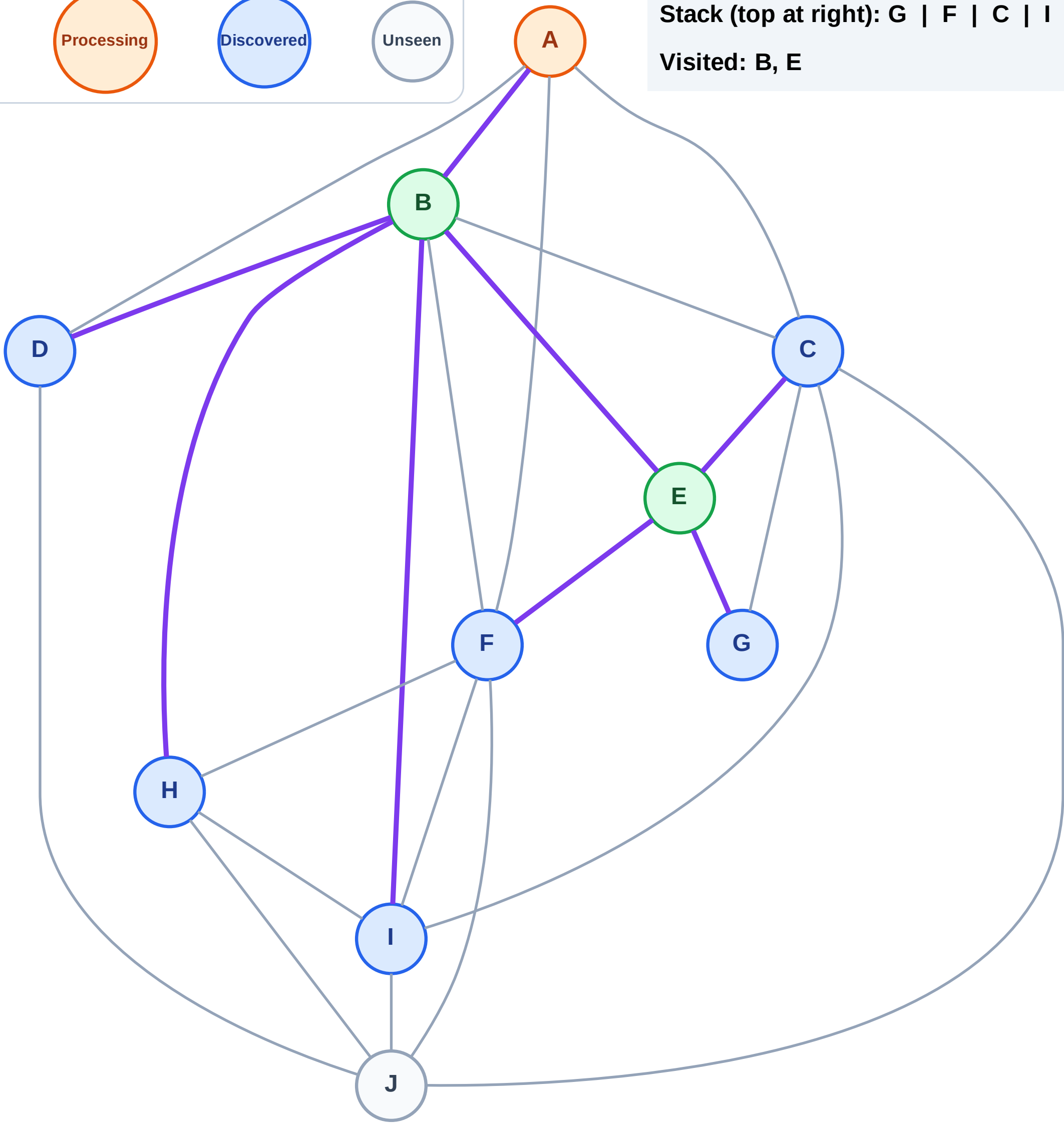
Complete

Processing

Discovered

Unseen

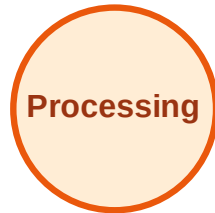
Stack (top at right): G | F | C | I | H | D
Visited: B, E



DFS Traversal

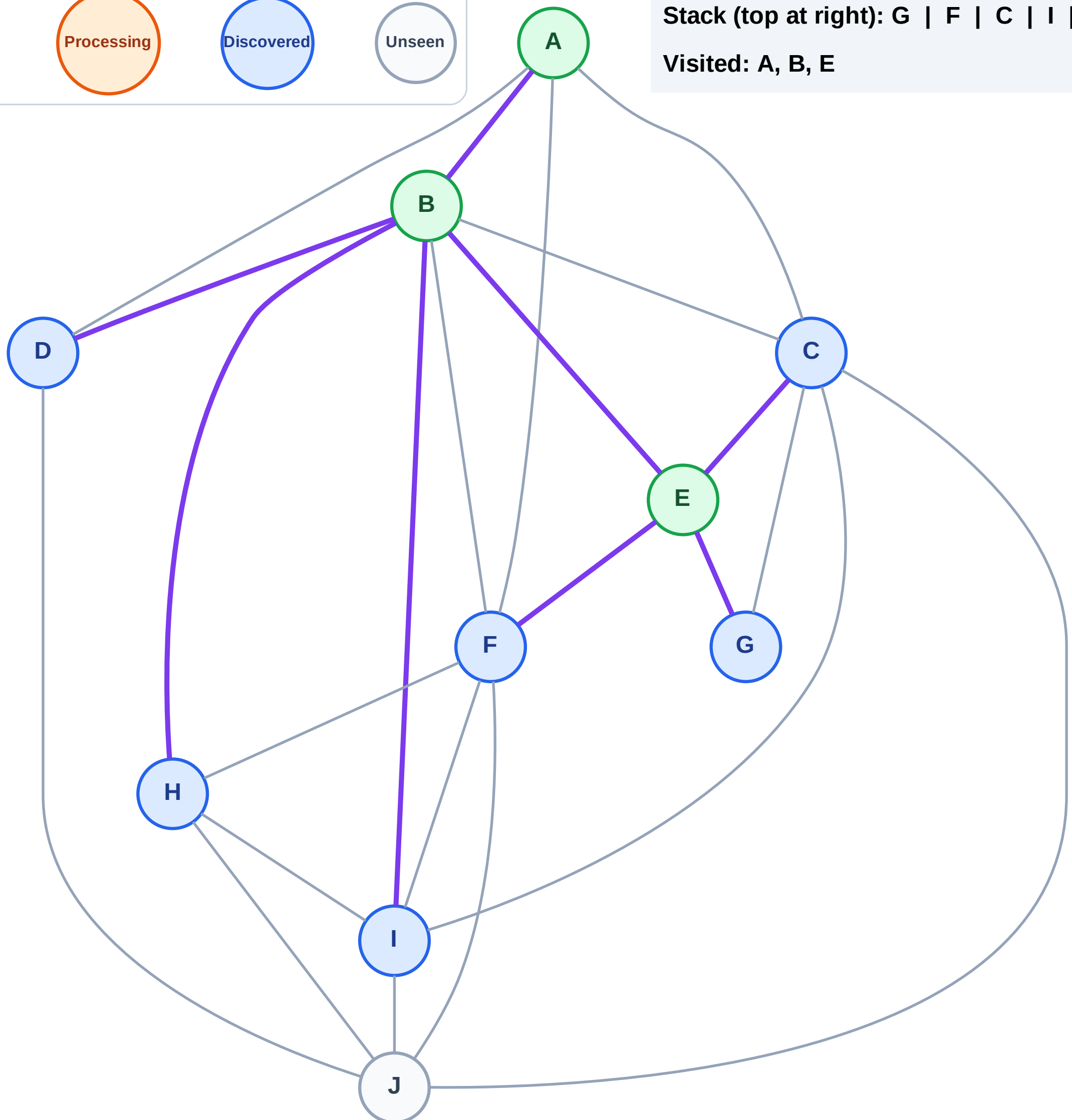
Step 7: No new neighbors from A

Legend



Stack (top at right): G | F | C | I | H | D

Visited: A, B, E



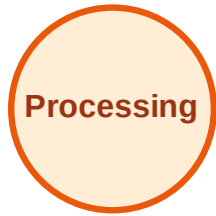
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



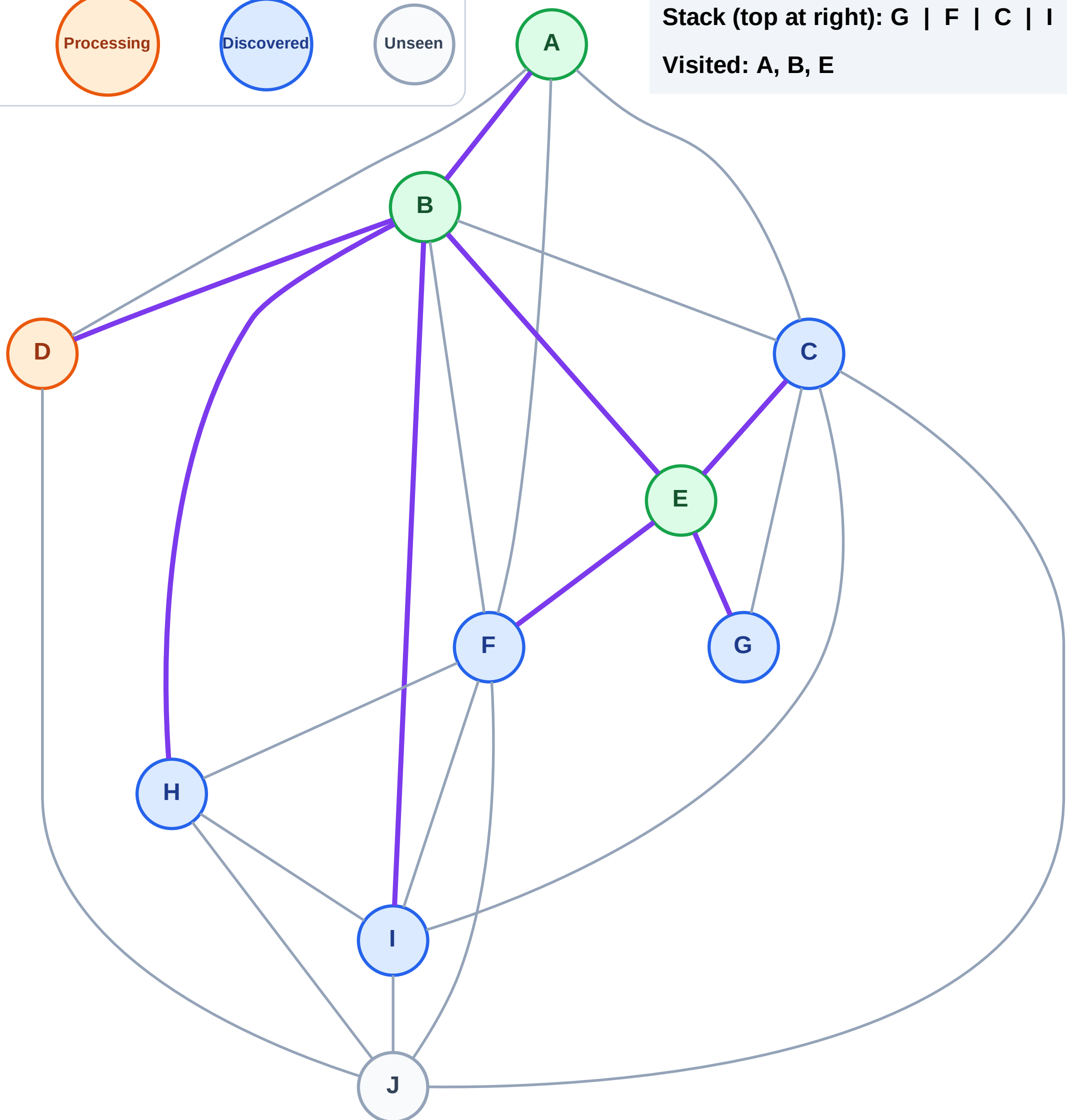
Discovered



Unseen

Stack (top at right): G | F | C | I | H

Visited: A, B, E



DFS Traversal

Step 9: Discover J from D

Legend

Complete

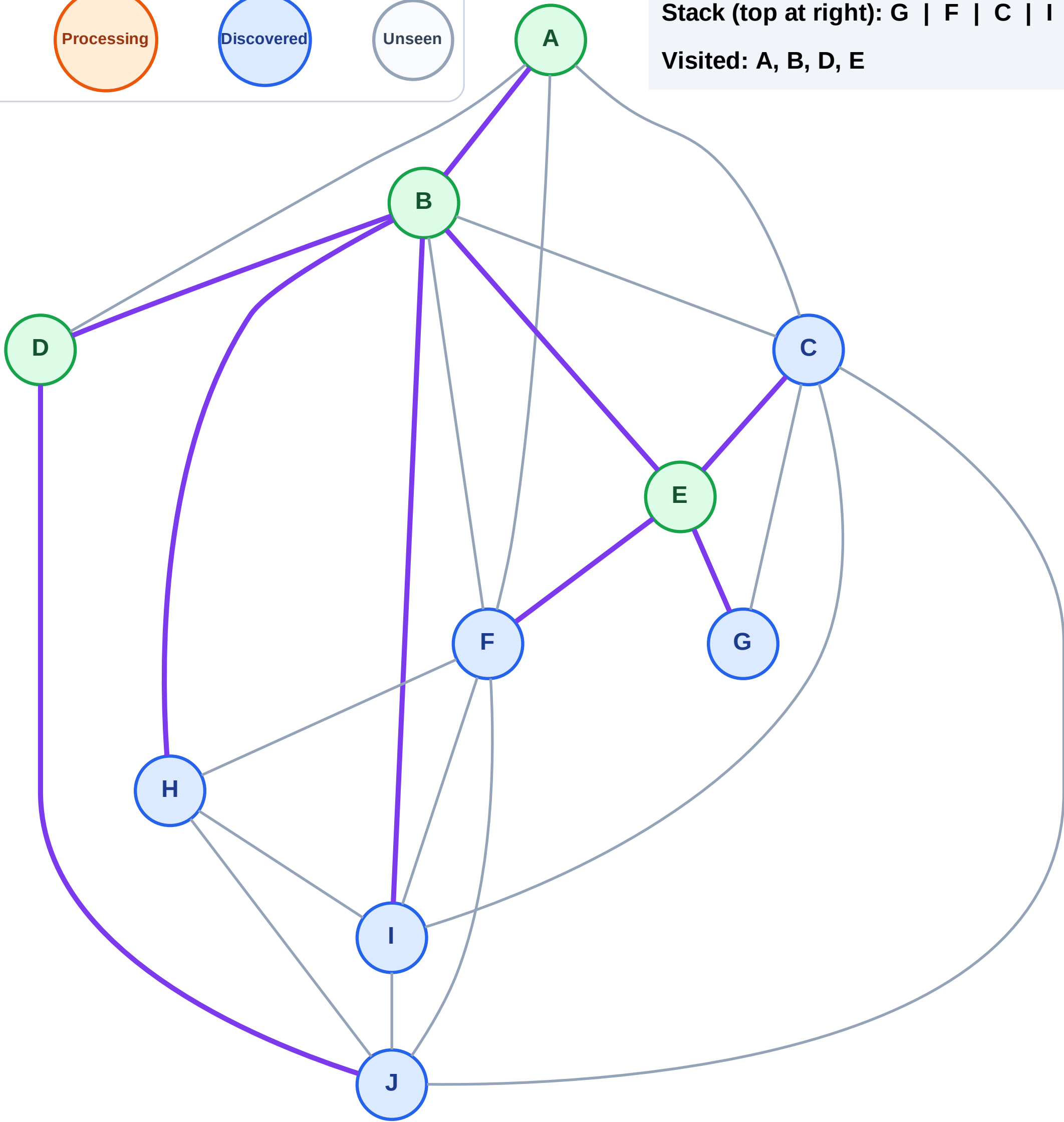
Processing

Discovered

Unseen

Stack (top at right): G | F | C | I | H | J

Visited: A, B, D, E



DFS Traversal

Step 10: Process node J

Legend

Complete

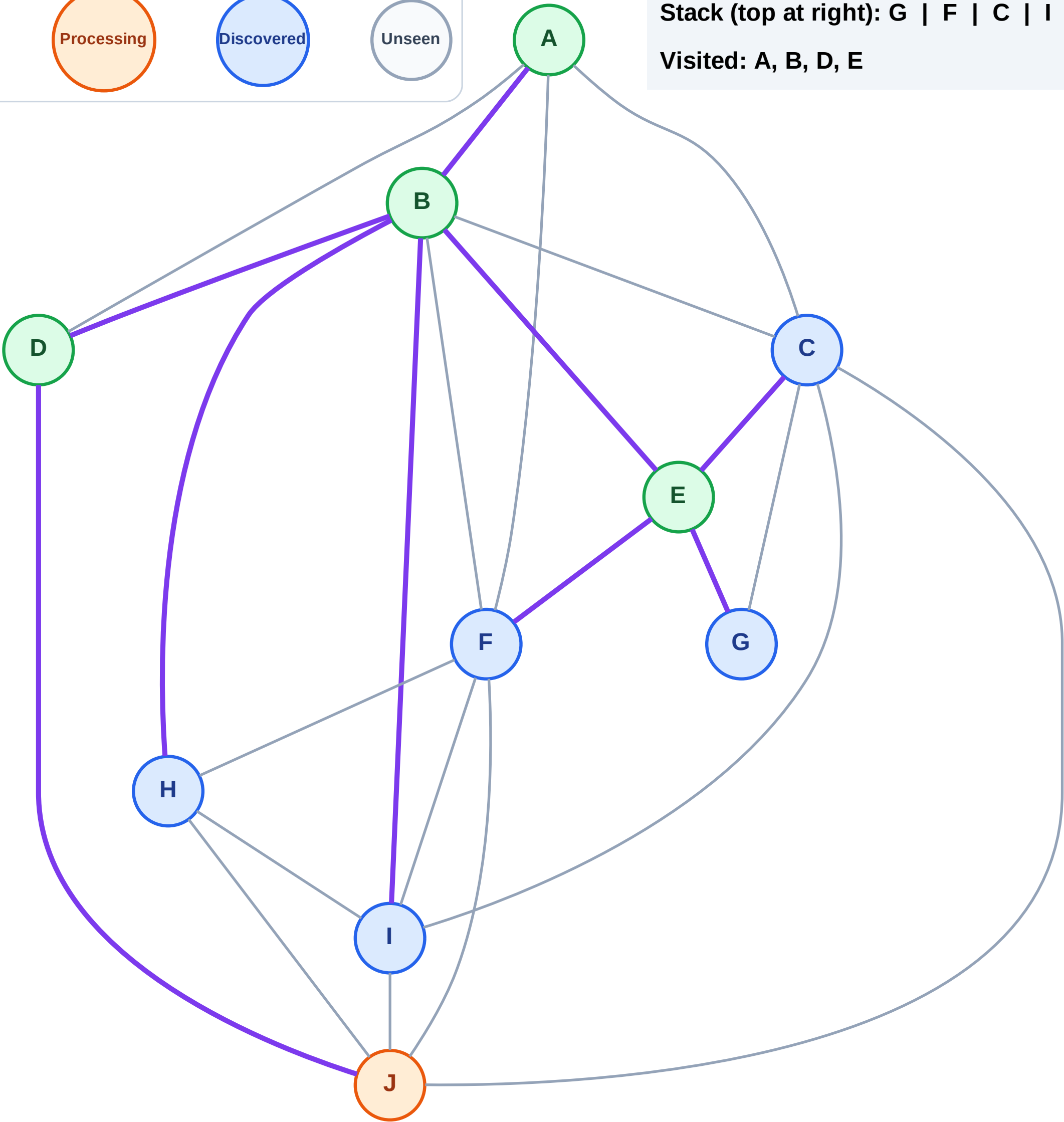
Processing

Discovered

Unseen

Stack (top at right): G | F | C | I | H

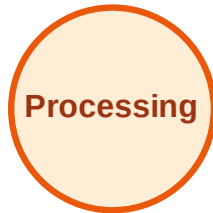
Visited: A, B, D, E



DFS Traversal

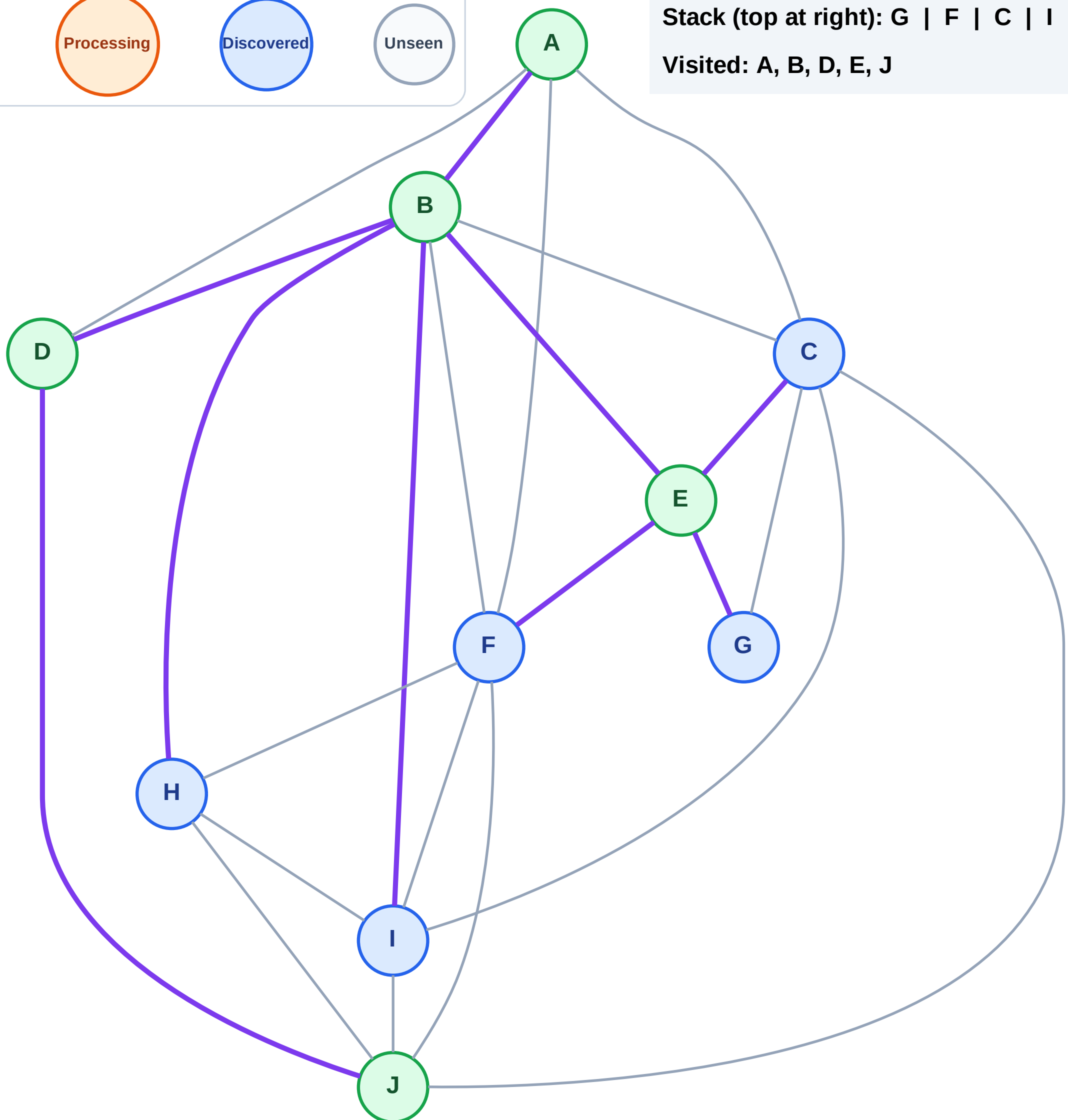
Step 11: No new neighbors from J

Legend



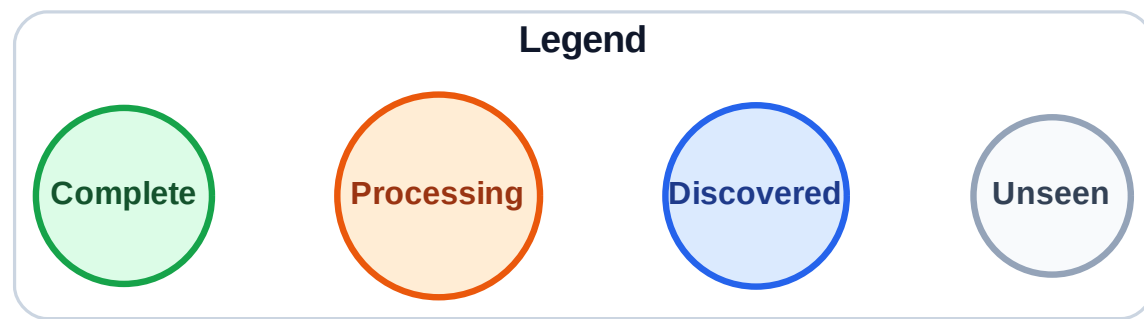
Stack (top at right): G | F | C | I | H

Visited: A, B, D, E, J

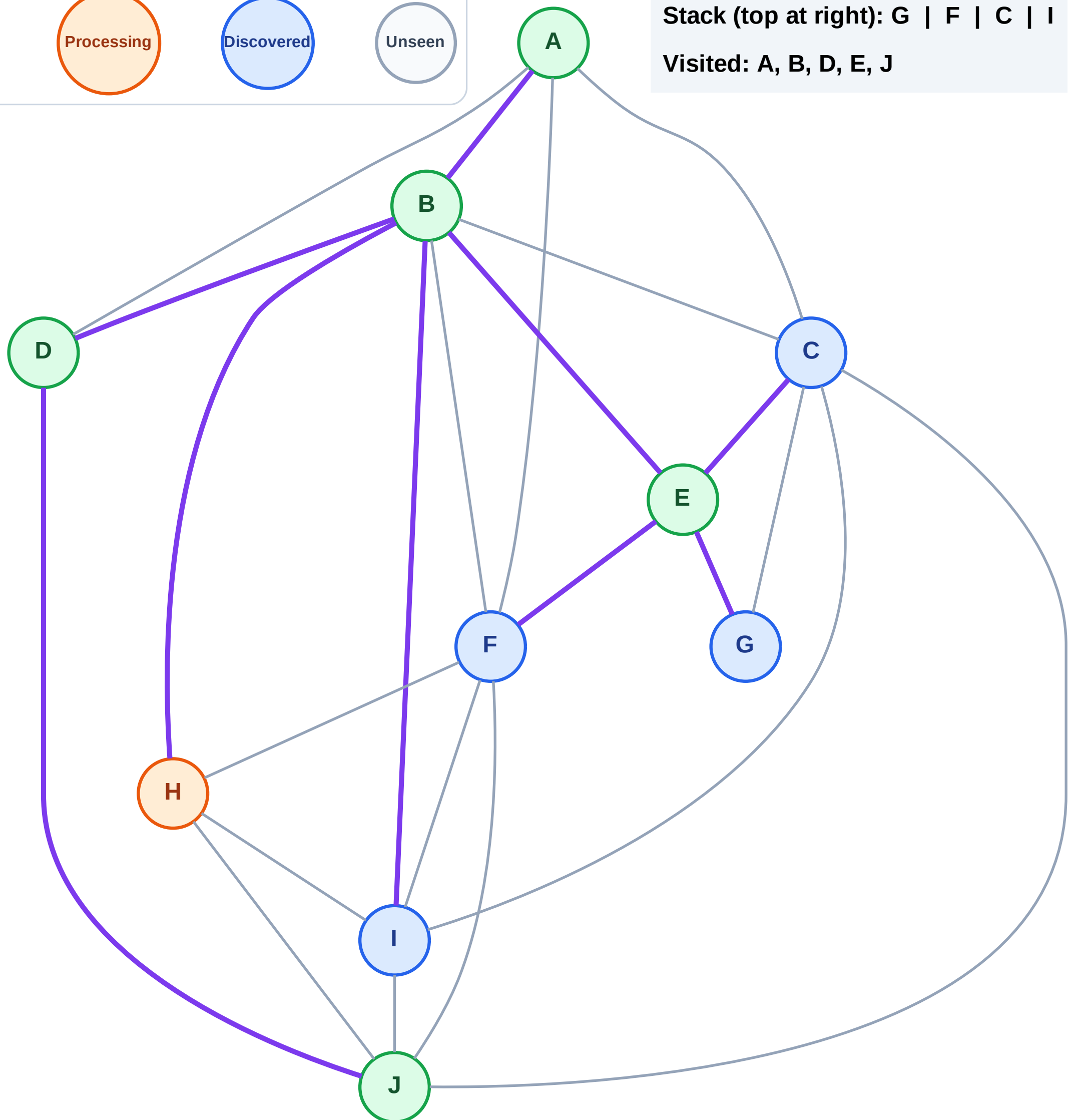


DFS Traversal

Step 12: Process node H



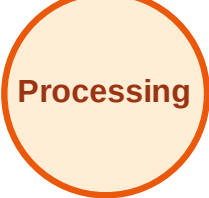
Stack (top at right): G | F | C | I
Visited: A, B, D, E, J



DFS Traversal

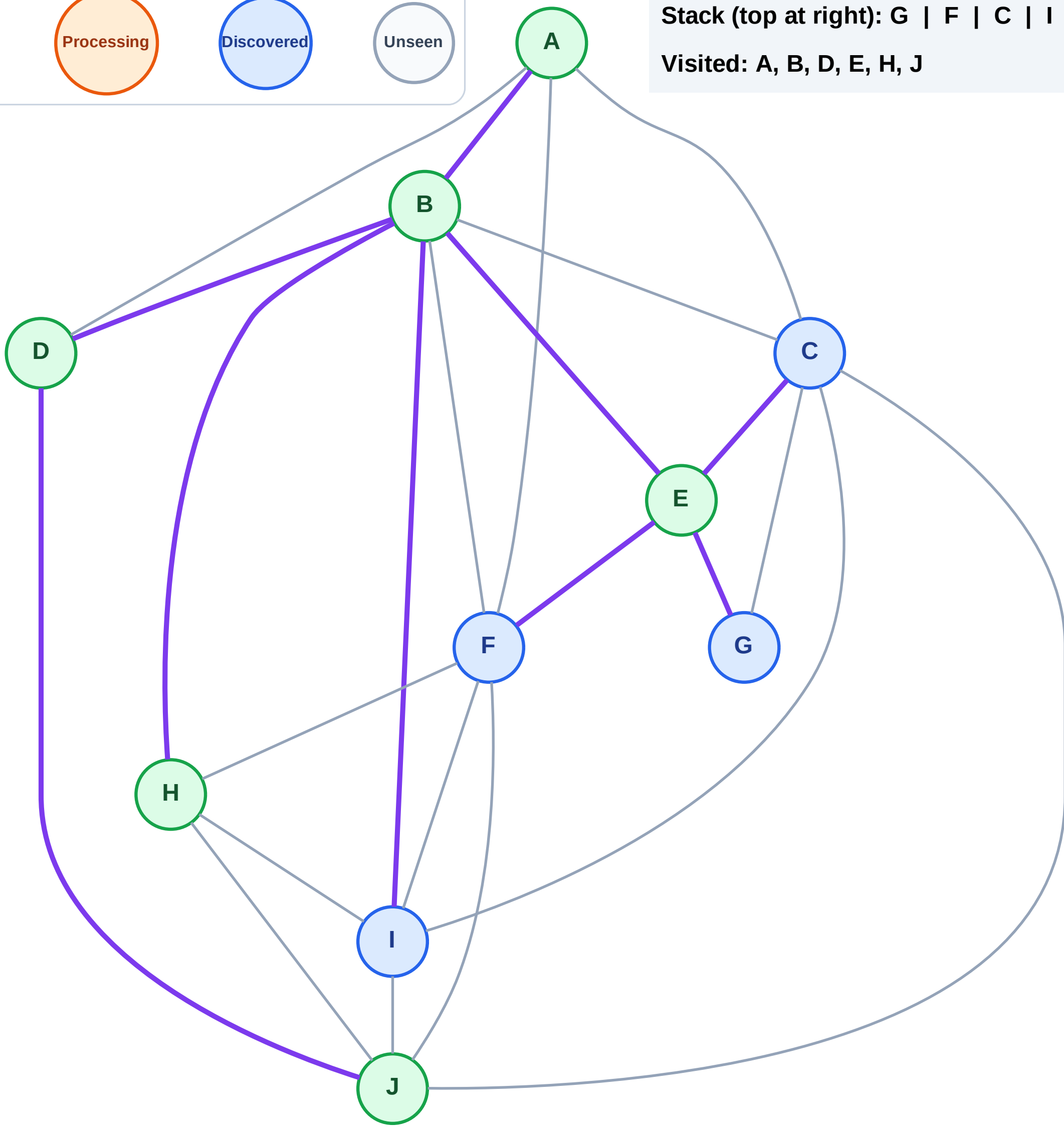
Step 13: No new neighbors from H

Legend



Stack (top at right): G | F | C | I

Visited: A, B, D, E, H, J



DFS Traversal

Step 14: Process node I

Legend

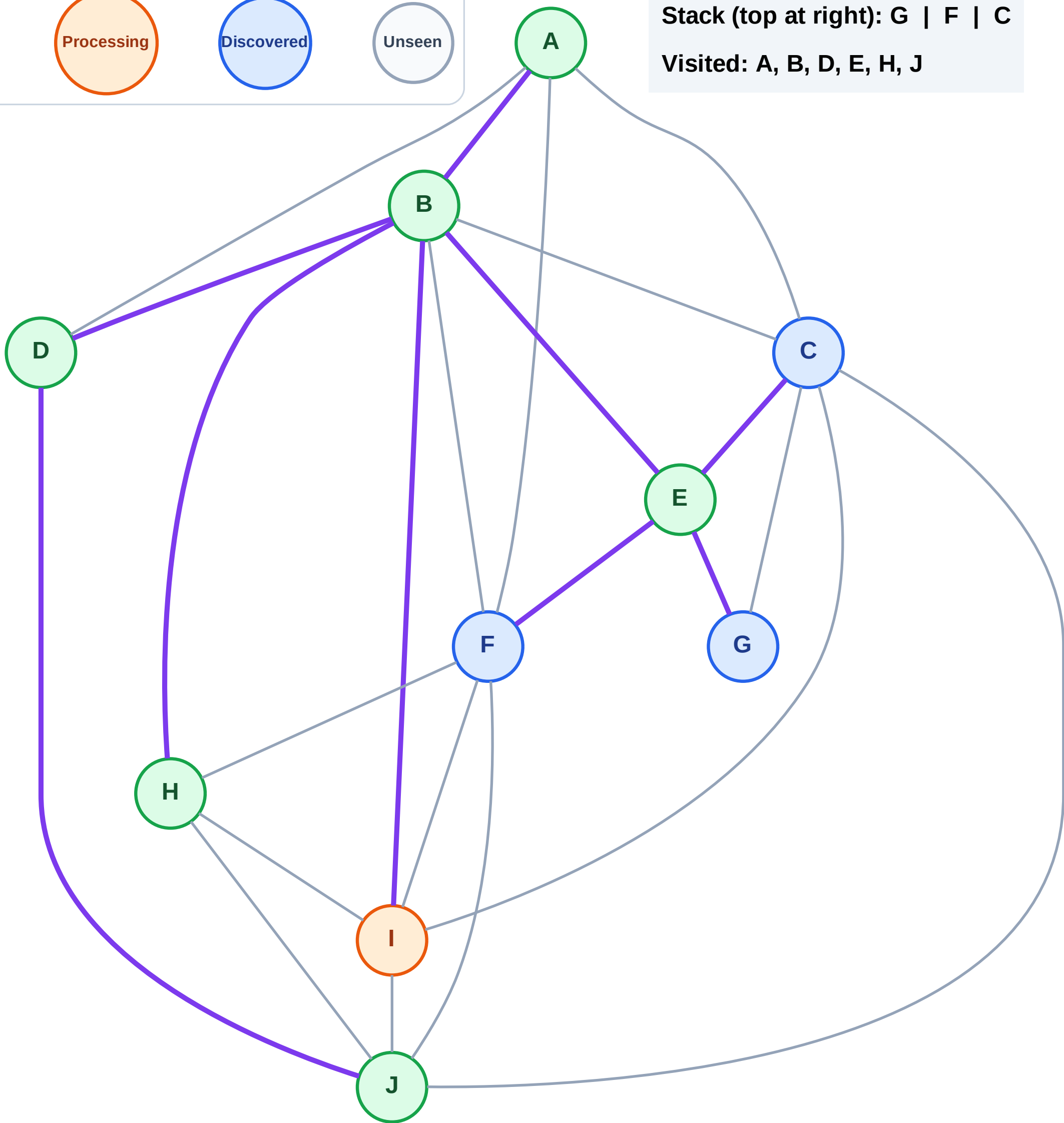
Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | C
Visited: A, B, D, E, H, J



DFS Traversal

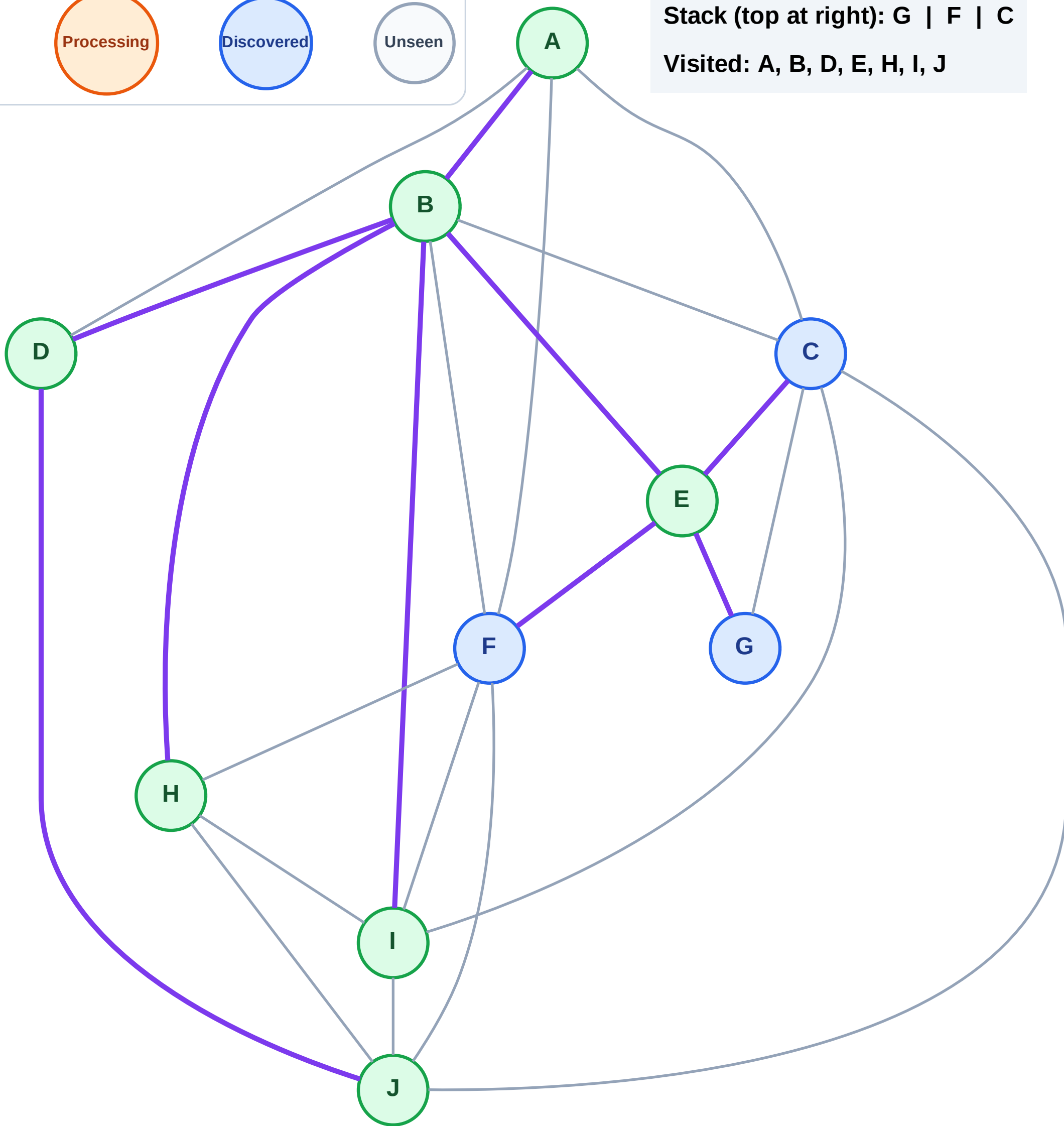
Step 15: No new neighbors from I

Legend



Stack (top at right): G | F | C

Visited: A, B, D, E, H, I, J



DFS Traversal

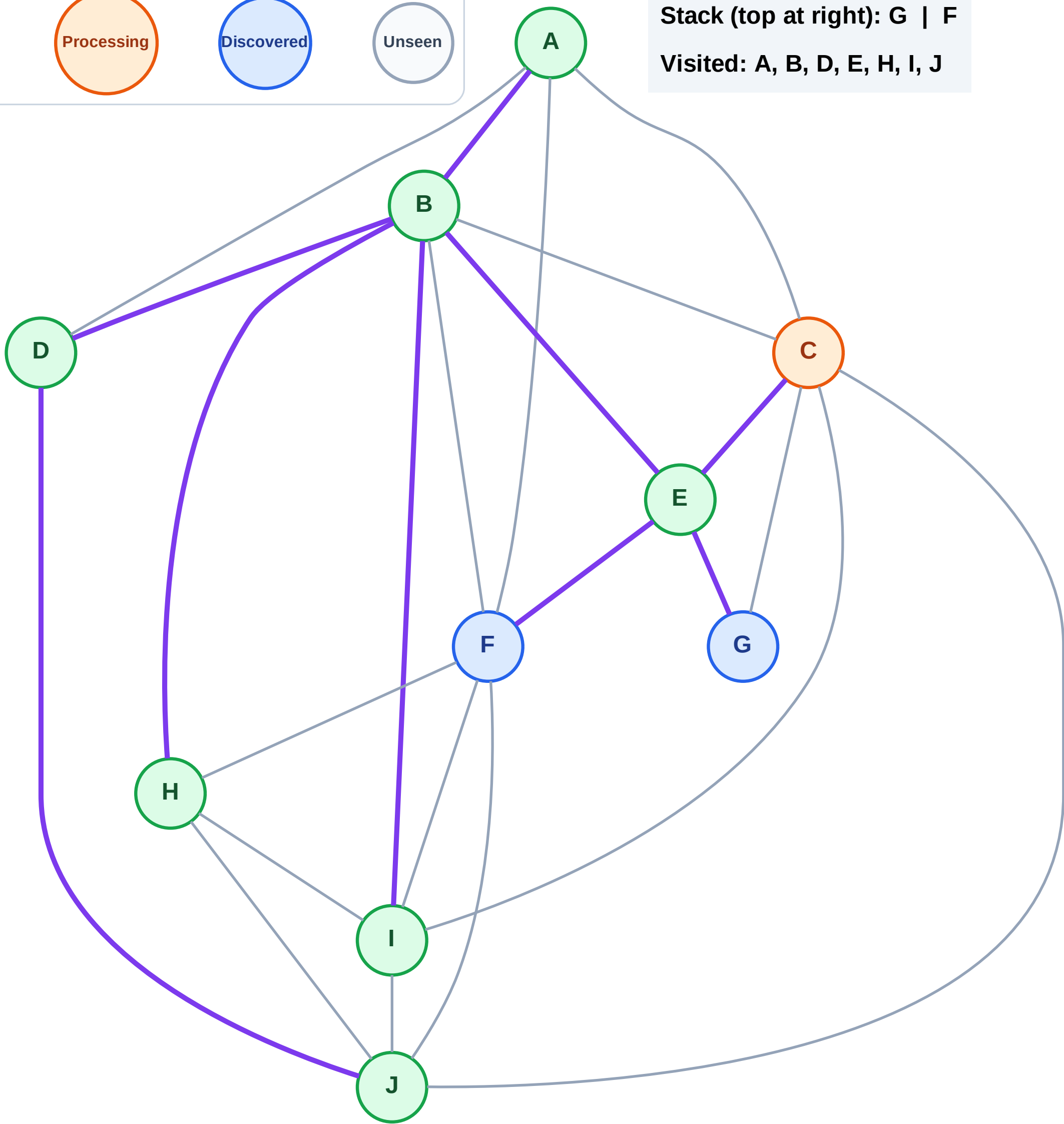
Step 16: Process node C

Legend



Stack (top at right): G | F

Visited: A, B, D, E, H, I, J



DFS Traversal

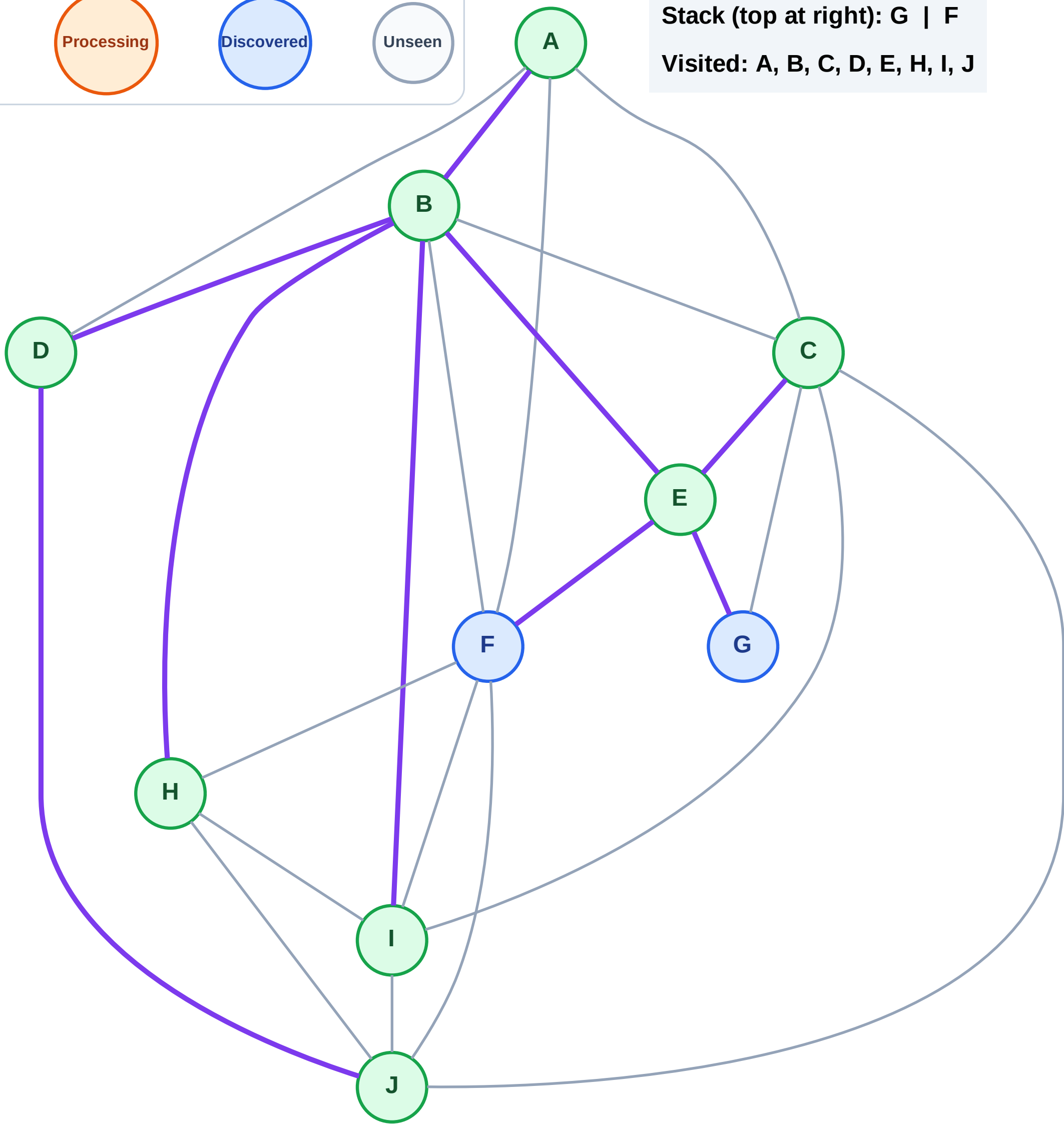
Step 17: No new neighbors from C

Legend



Stack (top at right): G | F

Visited: A, B, C, D, E, H, I, J



DFS Traversal

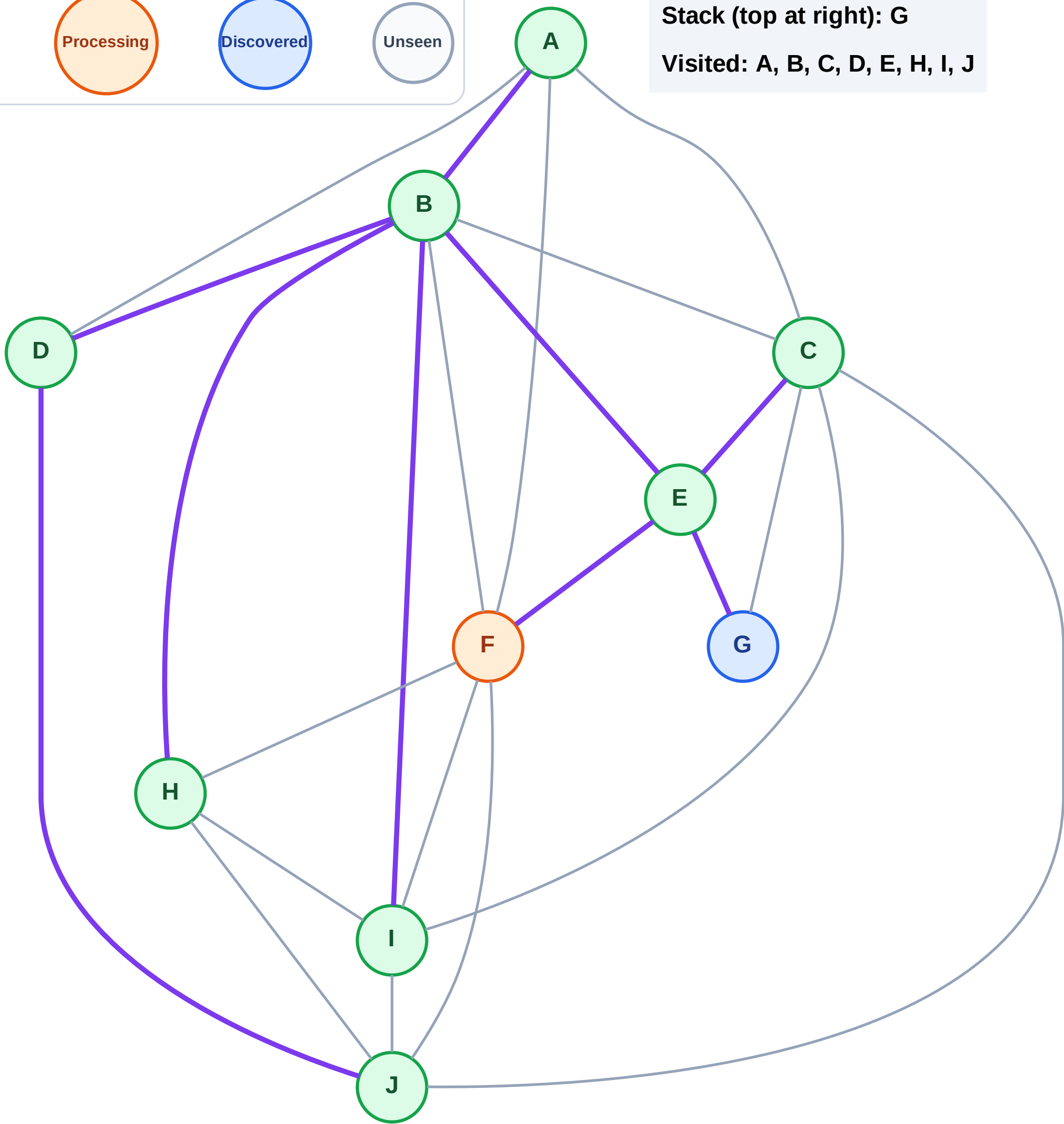
Step 18: Process node F

Legend



Stack (top at right): G

Visited: A, B, C, D, E, H, I, J



DFS Traversal

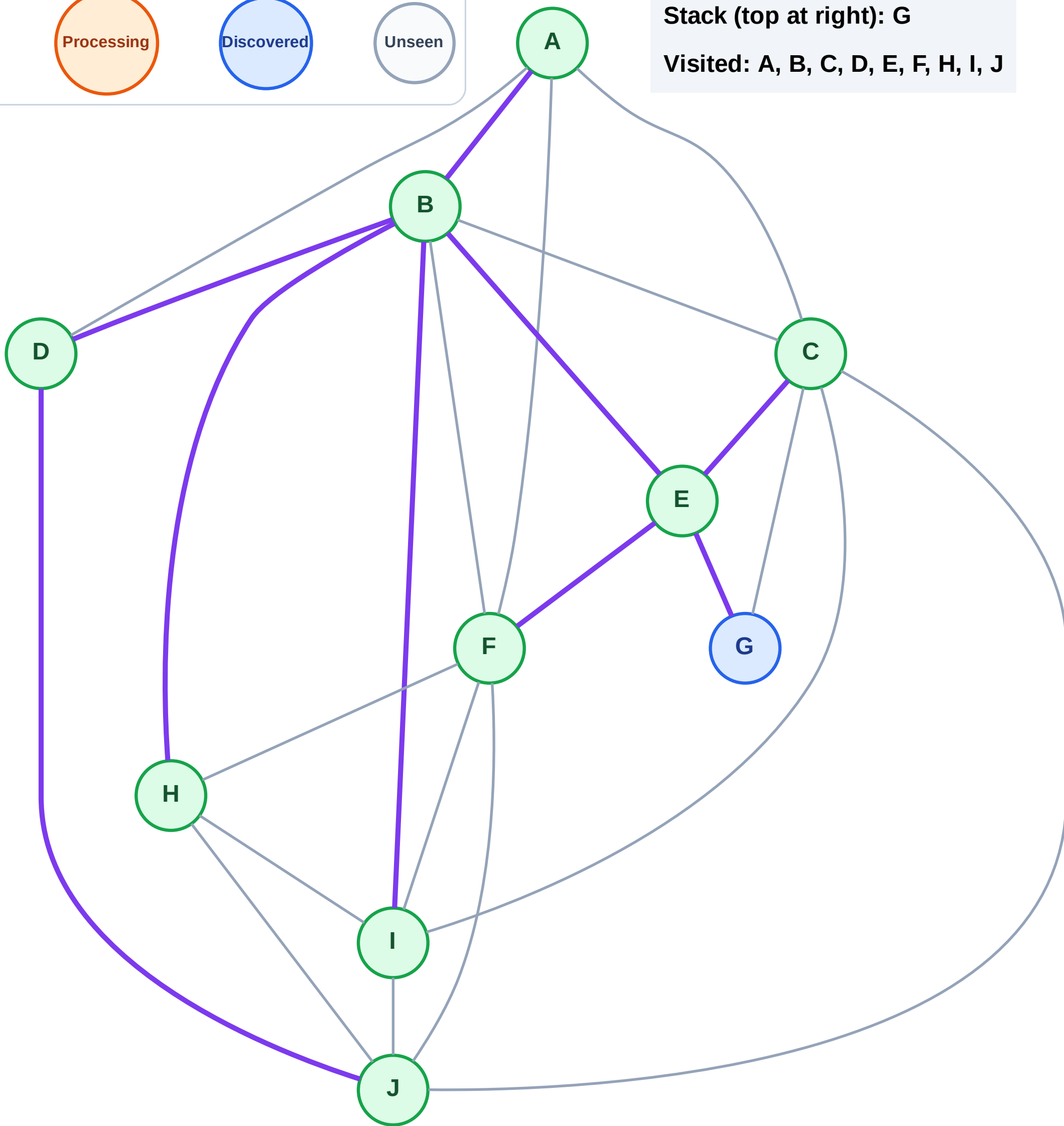
Step 19: No new neighbors from F

Legend



Stack (top at right): G

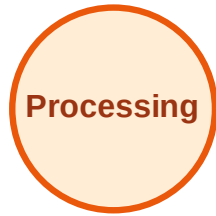
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

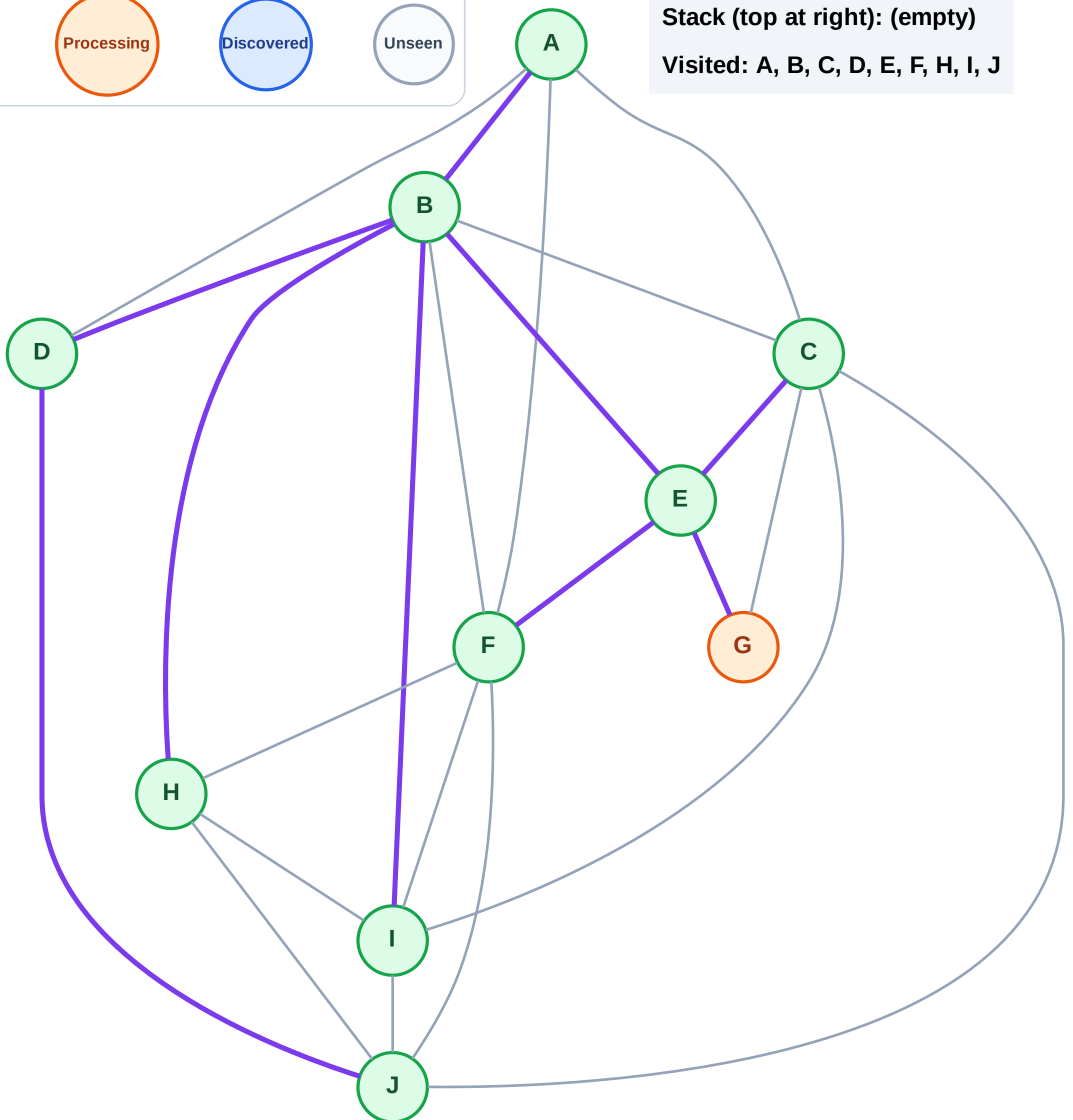
Step 20: Process node G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

